

Kinora Lumbar Spine

Clinical AI Orientation (Part 5)

Structured reference for RAG / AI-assisted physiotherapy consultation

Lumbar Anatomy, Biomechanics, Pathology, Red Flags & Rehabilitation

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

*This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established musculoskeletal medicine concepts from standard anatomical texts (Gray's, Moore, Netter, Bogduk), orthopedic assessment references (Magee), and clinical guidelines (NICE NG59, APTA Low Back Pain CPG) but must be verified against current peer-reviewed literature, local protocols, and individual patient presentation. **CAUDA EQUINA SYNDROME and all red flags require URGENT emergency medical/surgical referral — do not delay imaging or specialist care for conservative trial.** Urinary retention, saddle anesthesia, bilateral leg weakness, or progressive neurological deficit are emergencies. Diagnostic sensitivity/specificity values are approximate and vary by study.*

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1. Lumbar Vertebrae (L1-L5)

RECORD: L1 Vertebra

ID: VERT-L-01

Vertebral Level: L1

Name: First lumbar vertebra

Region: Lumbar spine — thoracolumbar transition

Bone Type: Irregular vertebra (lumbar type)

Body: Kidney-shaped body; smaller than L2; anterior height slightly greater than posterior; endplates flat

Pedicles: L1 pedicles short and strong; project posterolaterally from body; form lateral vertebral canal boundary; pedicle screw trajectory target in surgery; nerve root exits below pedicle at same numbered level (e.g., L4 root under L4 pedicle)

Lamina: Broad laminae unite spinous process to articular pillars; ligamentum flavum attaches to anterior surface; multifidus origin from lamina and spinous process; laminectomy removes lamina for decompression

Spinous Process: Broad quadrilateral spinous process; horizontal orientation (vs thoracic oblique); supraspinous and interspinous ligament attachment; palpable midline landmark; L1 SP palpable with patient flexed

Transverse Processes: Long slender transverse processes; no accessory process; mamillary process on superior articular process

Pars Interarticularis: Pars interarticularis (pedicle-lamina junction) between superior and inferior articular processes; stress concentration site for spondylolysis; L5 pars most commonly affected in athletes; orientation permits sagittal plane motion at inferior facets

Articular Facets: Superior articular facets face medially-posteriorly; inferior facets face laterally-anteriorly; paired zygapophyseal joints with L12 superiorly and L2 inferiorly; hyaline cartilage; synovial joint

Facet Orientation: ~90 deg coronal (frontal plane dominant)

Vertebral Canal: Triangular canal containing cauda equina (below ~L1-L2 conus); dura mater, arachnoid, CSF; epidural space; ligamentum flavum forms posterior wall; stenosis reduces cross-sectional area

Intervertebral Foramina: Formed by pedicles superior/inferior, vertebral body/disc anteriorly, facet joints posteriorly; L1 nerve root exits foramen; far-lateral/foraminal disc herniation can compress exiting root; foraminal height decreases in extension and disc height loss

Articulations: L1-L2 disc (symphysis); bilateral facet joints; T12-L1 junction above L1

Muscle Attachments: Multifidus (SP, lamina, mamillary); erector spinae — iliocostalis lumborum, longissimus thoracis (TP, angles); quadratus lumborum (TP L1-L4, iliolumbar); psoas major (bodies/Tps L1-L5); latissimus dorsi via thoracolumbar fascia; intertransversarii and rotatores segmental

Ligament Attachments: ALL on anterior body; PLL posterior body; ligamentum flavum lamina-lamina; interspinous and supraspinous between SPs; intertransverse between TPs; intertransverse ligaments

Blood Supply: Segmental lumbar arteries from aorta (typically 4 pairs); spinal branches through intervertebral foramen; metameric supply to body, pedicle, lamina; L1 radicular artery may contribute (variable; artery of Adamkiewicz usually T9-L2)

Venous Drainage: Segmental lumbar veins to inferior vena cava/ascending lumbar vein; internal vertebral venous plexus (Batson); valveless plexus permits metastatic spread; epidural venous engorgement in stenosis

Innervation: Segmental L1 medial branch dorsal rami innervate facet capsules and multifidus; L1 ventral ramus contributes to lumbar plexus branches; recurrent sinuvertebral nerve to disc/PLL

Biomechanical Role: L1 contributes to lordosis (~40-60 deg total lumbar); thoracolumbar transition stiffness at L1

Normal Motion: Segmental flexion 8-12 deg, extension 4-6 deg, lateral flexion 4-6 deg, rotation 1-2 deg (rotation minimal in lumbar); coupled motions; L4-L5 and L5-S1 greatest total ROM

Load Transmission: 80% axial load through anterior column (body/disc); 20% through facet joints at neutral standing (approximate); facets share more load in extension; disc bears compression; posterior elements resist shear

Clinical Importance: L1 common site for degenerative change, facet arthropathy, and disc pathology; upper lumbar disc herniation less common than L4-L5/L5-S1

Common Injuries: Disc herniation, facet joint sprain/arthropathy, compression fracture (osteoporosis), spondylolysis (L5>L4), spondylolisthesis, pars stress reaction, transverse process avulsion (rare)

Imaging: AP/lateral X-ray: alignment, spondylolisthesis Meyerding grade, disc height, lordosis; MRI: disc hydration (Pfirrmann), nerve root compression, Modic endplate changes, stenosis; CT: pars defects, facet tropism, surgical planning

Healing: Bone: 6-12 weeks uncomplicated fracture; pars non-union possible in spondylolysis; disc: limited intrinsic healing; annular repair fibrotic; modifiable via load management and rehab

Rehabilitation: Direction preference (flexion vs extension) per assessment; core stabilization; progressive loading; avoid prolonged bed rest; motor control multifidus/transversus; return to activity when pain controlled and functional goals met

References: Moore Clinically Oriented Anatomy; Bogduk Clinical Anatomy of the Lumbar Spine and Sacrum 2012; Netter Atlas; Gray's Anatomy

RECORD: L2 Vertebra

ID: VERT-L-02

Vertebral Level: L2

Name: Second lumbar vertebra

Region: Lumbar spine

Bone Type: Irregular vertebra (lumbar type)

Body: Large vertebral body; increased cross-sectional area vs L1; supports upper lumbar load

Pedicles: L2 pedicles short and strong; project posterolaterally from body; form lateral vertebral canal boundary; pedicle screw trajectory target in surgery; nerve root exits below pedicle at same numbered level (e.g., L4 root under L4 pedicle)

Lamina: Broad laminae unite spinous process to articular pillars; ligamentum flavum attaches to anterior surface; multifidus origin from lamina and spinous process; laminectomy removes lamina for decompression

Spinous Process: Broad quadrilateral spinous process; horizontal orientation (vs thoracic oblique); supraspinous and interspinous ligament attachment; palpable midline landmark; L2 SP palpable with patient flexed

Transverse Processes: Long TP; mamillary processes present; attachment for lumbar fascia and quadratus lumborum

Pars Interarticularis: Pars interarticularis (pedicle-lamina junction) between superior and inferior articular processes; stress concentration site for spondylolysis; L5 pars most commonly affected in athletes; orientation permits sagittal plane motion at inferior facets

Articular Facets: Superior articular facets face medially-posteriorly; inferior facets face laterally-anteriorly; paired zygapophyseal joints with L1 superiorly and L3 inferiorly; hyaline cartilage; synovial joint

Facet Orientation: ~80-85 deg coronal

Vertebral Canal: Triangular canal containing cauda equina (below ~L1-L2 conus); dura mater, arachnoid, CSF; epidural space; ligamentum flavum forms posterior wall; stenosis reduces cross-sectional area

Intervertebral Foramina: Formed by pedicles superior/inferior, vertebral body/disc anteriorly, facet joints posteriorly; L2 nerve root exits foramen; far-lateral/foraminal disc herniation can compress exiting root; foraminal height decreases in extension and disc height loss

Articulations: L2-L3 disc (symphysis); bilateral facet joints; L1-L2 and L2-L3 facet pairs

Muscle Attachments: Multifidus (SP, lamina, mamillary); erector spinae — iliocostalis lumborum, longissimus thoracis (TP, angles); quadratus lumborum (TP L1-L4, iliolumbar); psoas major (bodies/Tps L1-L5); latissimus dorsi via thoracolumbar fascia; intertransversarii and rotatores segmental

Ligament Attachments: ALL on anterior body; PLL posterior body; ligamentum flavum lamina-lamina; interspinous and supraspinous between SPs; intertransverse between TPs; intertransverse ligaments

Blood Supply: Segmental lumbar arteries from aorta (typically 4 pairs); spinal branches through intervertebral foramen; metameric supply to body, pedicle, lamina; L2 radicular artery may contribute (variable; artery of Adamkiewicz usually T9-L2)

Venous Drainage: Segmental lumbar veins to inferior vena cava/ascending lumbar vein; internal vertebral venous plexus (Batson); valveless plexus permits metastatic spread; epidural venous engorgement in stenosis

Innervation: Segmental L2 medial branch dorsal rami innervate facet capsules and multifidus; L2 ventral ramus contributes to lumbar plexus branches; recurrent sinuvertebral nerve to disc/PLL

Biomechanical Role: L2 contributes to lordosis (~40-60 deg total lumbar); central lumbar load bearing and flexion-extension

Normal Motion: Segmental flexion 8-12 deg, extension 4-6 deg, lateral flexion 4-6 deg, rotation 1-2 deg (rotation minimal in lumbar); coupled motions; L4-L5 and L5-S1 greatest total ROM

Load Transmission: 80% axial load through anterior column (body/disc); 20% through facet joints at neutral standing (approximate); facets share more load in extension; disc bears compression; posterior elements resist shear

Clinical Importance: L2 common site for degenerative change, facet arthropathy, and disc pathology; upper lumbar disc herniation less common than L4-L5/L5-S1

Common Injuries: Disc herniation, facet joint sprain/arthropathy, compression fracture (osteoporosis), spondylolysis (L5>L4), spondylolisthesis, pars stress reaction, transverse process avulsion (rare)

Imaging: AP/lateral X-ray: alignment, spondylolisthesis Meyerding grade, disc height, lordosis; MRI: disc hydration (Pfirrmann), nerve root compression, Modic endplate changes, stenosis; CT: pars defects, facet tropism, surgical planning

Healing: Bone: 6-12 weeks uncomplicated fracture; pars non-union possible in spondylolysis; disc: limited intrinsic healing; annular repair fibrotic; modifiable via load management and rehab

Rehabilitation: Direction preference (flexion vs extension) per assessment; core stabilization; progressive loading; avoid prolonged bed rest; motor control multifidus/transversus; return to activity when pain controlled and functional goals met

References: Moore Clinically Oriented Anatomy; Bogduk Clinical Anatomy of the Lumbar Spine and Sacrum 2012; Netter Atlas; Gray's Anatomy

RECORD: L3 Vertebra

ID: VERT-L-03

Vertebral Level: L3

Name: Third lumbar vertebra

Region: Lumbar spine

Bone Type: Irregular vertebra (lumbar type)

Body: Largest lumbar body typically; central load-bearing segment; endplates broad

Pedicles: L3 pedicles short and strong; project posterolaterally from body; form lateral vertebral canal boundary; pedicle screw trajectory target in surgery; nerve root exits below pedicle at same numbered level (e.g., L4 root under L4 pedicle)

Lamina: Broad laminae unite spinous process to articular pillars; ligamentum flavum attaches to anterior surface; multifidus origin from lamina and spinous process; laminectomy removes lamina for decompression

Spinous Process: Broad quadrilateral spinous process; horizontal orientation (vs thoracic oblique); supraspinous and interspinous ligament attachment; palpable midline landmark; L3 SP palpable with patient flexed

Transverse Processes: Long TP; iliocostalis and longissimus attachments; intertransverse ligament anchor

Pars Interarticularis: Pars interarticularis (pedicle-lamina junction) between superior and inferior articular processes; stress concentration site for spondylolysis; L5 pars most commonly affected in athletes; orientation permits sagittal plane motion at inferior facets

Articular Facets: Superior articular facets face medially-posteriorly; inferior facets face laterally-anteriorly; paired zygapophyseal joints with L2 superiorly and L4 inferiorly; hyaline cartilage; synovial joint

Facet Orientation: ~70-75 deg coronal

Vertebral Canal: Triangular canal containing cauda equina (below ~L1-L2 conus); dura mater, arachnoid, CSF; epidural space; ligamentum flavum forms posterior wall; stenosis reduces cross-sectional area

Intervertebral Foramina: Formed by pedicles superior/inferior, vertebral body/disc anteriorly, facet joints posteriorly; L3 nerve root exits foramen; far-lateral/foraminal disc herniation can compress exiting root; foraminal height decreases in extension and disc height loss

Articulations: L3-L4 disc (symphysis); bilateral facet joints; L2-L3 and L3-L4 facet pairs

Muscle Attachments: Multifidus (SP, lamina, mamillary); erector spinae — iliocostalis lumborum, longissimus thoracis (TP, angles); quadratus lumborum (TP L1-L4, iliolumbar); psoas major (bodies/Tps L1-L5); latissimus dorsi via thoracolumbar fascia; intertransversarii and rotatores segmental

Ligament Attachments: ALL on anterior body; PLL posterior body; ligamentum flavum lamina-lamina; interspinous and supraspinous between SPs; intertransverse between TPs; intertransverse ligaments

Blood Supply: Segmental lumbar arteries from aorta (typically 4 pairs); spinal branches through intervertebral foramen; metameric supply to body, pedicle, lamina; L3 radicular artery may contribute (variable; artery of Adamkiewicz usually T9-L2)

Venous Drainage: Segmental lumbar veins to inferior vena cava/ascending lumbar vein; internal vertebral venous plexus (Batson); valveless plexus permits metastatic spread; epidural venous engorgement in stenosis

Innervation: Segmental L3 medial branch dorsal rami innervate facet capsules and multifidus; L3 ventral ramus contributes to lumbar plexus branches; recurrent sinuvertebral nerve to disc/PLL

Biomechanical Role: L3 contributes to lordosis (~40-60 deg total lumbar); central lumbar load bearing and flexion-extension

Normal Motion: Segmental flexion 8-12 deg, extension 4-6 deg, lateral flexion 4-6 deg, rotation 1-2 deg (rotation minimal in lumbar); coupled motions; L4-L5 and L5-S1 greatest total ROM

Load Transmission: 80% axial load through anterior column (body/disc); 20% through facet joints at neutral standing (approximate); facets share more load in extension; disc bears compression; posterior elements resist shear

Clinical Importance: L3 common site for degenerative change, facet arthropathy, and disc pathology; referral patterns for radiculopathy at this level

Common Injuries: Disc herniation, facet joint sprain/arthropathy, compression fracture (osteoporosis), spondylolysis (L5>L4), spondylolisthesis, pars stress reaction, transverse process avulsion (rare)

Imaging: AP/lateral X-ray: alignment, spondylolisthesis Meyerding grade, disc height, lordosis; MRI: disc hydration (Pfirrmann), nerve root compression, Modic endplate changes, stenosis; CT: pars defects, facet tropism, surgical planning

Healing: Bone: 6-12 weeks uncomplicated fracture; pars non-union possible in spondylolysis; disc: limited intrinsic healing; annular repair fibrotic; modifiable via load management and rehab

Rehabilitation: Direction preference (flexion vs extension) per assessment; core stabilization; progressive loading; avoid prolonged bed rest; motor control multifidus/transversus; return to activity when pain controlled and functional goals met

References: Moore Clinically Oriented Anatomy; Bogduk Clinical Anatomy of the Lumbar Spine and Sacrum 2012; Netter Atlas; Gray's Anatomy

RECORD: L4 Vertebra

ID: VERT-L-04

Vertebral Level: L4

Name: Fourth lumbar vertebra

Region: Lumbar spine

Bone Type: Irregular vertebra (lumbar type)

Body: Large body; iliolumbar ligament attachment region on transverse processes; transition to lumbosacral mechanics

Pedicles: L4 pedicles short and strong; project posterolaterally from body; form lateral vertebral canal boundary; pedicle screw trajectory target in surgery; nerve root exits below pedicle at same numbered level (e.g., L4 root under L4 pedicle)

Lamina: Broad laminae unite spinous process to articular pillars; ligamentum flavum attaches to anterior surface; multifidus origin from lamina and spinous process; laminectomy removes lamina for decompression

Spinous Process: Broad quadrilateral spinous process; horizontal orientation (vs thoracic oblique); supraspinous and interspinous ligament attachment; palpable midline landmark; L4 SP palpable with patient flexed

Transverse Processes: TP with iliolumbar ligament attachment (L4 strongest iliolumbar origin); QL insertion

Pars Interarticularis: Pars interarticularis (pedicle-lamina junction) between superior and inferior articular processes; stress concentration site for spondylolysis; L5 pars most commonly affected in athletes; orientation permits sagittal plane motion at

inferior facets

Articular Facets: Superior articular facets face medially-posteriorly; inferior facets face laterally-anteriorly; paired zygapophyseal joints with L3 superiorly and L5 inferiorly; hyaline cartilage; synovial joint

Facet Orientation: ~60-65 deg (transitional toward sagittal)

Vertebral Canal: Triangular canal containing cauda equina (below ~L1-L2 conus); dura mater, arachnoid, CSF; epidural space; ligamentum flavum forms posterior wall; stenosis reduces cross-sectional area

Intervertebral Foramina: Formed by pedicles superior/inferior, vertebral body/disc anteriorly, facet joints posteriorly; L4 nerve root exits foramen; far-lateral/foraminal disc herniation can compress exiting root; foraminal height decreases in extension and disc height loss

Articulations: L4-L5 disc (symphysis); bilateral facet joints; L3-L4 and L4-L5 facet pairs

Muscle Attachments: Multifidus (SP, lamina, mamillary); erector spinae — iliocostalis lumborum, longissimus thoracis (TP, angles); quadratus lumborum (TP L1-L4, iliolumbar); psoas major (bodies/Tps L1-L5); latissimus dorsi via thoracolumbar fascia; intertransversarii and rotatores segmental

Ligament Attachments: ALL on anterior body; PLL posterior body; ligamentum flavum lamina-lamina; interspinous and supraspinous between SPs; intertransverse between TPs; iliolumbar ligament attachment on TP (strongest L4-L5)

Blood Supply: Segmental lumbar arteries from aorta (typically 4 pairs); spinal branches through intervertebral foramen; metameric supply to body, pedicle, lamina; L4 radicular artery may contribute (variable; artery of Adamkiewicz usually T9-L2)

Venous Drainage: Segmental lumbar veins to inferior vena cava/ascending lumbar vein; internal vertebral venous plexus (Batson); valveless plexus permits metastatic spread; epidural venous engorgement in stenosis

Innervation: Segmental L4 medial branch dorsal rami innervate facet capsules and multifidus; L4 ventral ramus contributes to lumbar plexus branches; recurrent sinuvertebral nerve to disc/PLL

Biomechanical Role: L4 contributes to lordosis (~40-60 deg total lumbar); central lumbar load bearing and flexion-extension

Normal Motion: Segmental flexion 8-12 deg, extension 4-6 deg, lateral flexion 4-6 deg, rotation 1-2 deg (rotation minimal in lumbar); coupled motions; L4-L5 and L5-S1 greatest total ROM

Load Transmission: 80% axial load through anterior column (body/disc); 20% through facet joints at neutral standing (approximate); facets share more load in extension; disc bears compression; posterior elements resist shear

Clinical Importance: L4 common site for degenerative change, facet arthropathy, and disc pathology; referral patterns for radiculopathy at this level

Common Injuries: Disc herniation, facet joint sprain/arthropathy, compression fracture (osteoporosis), spondylolysis (L5>L4), spondylolisthesis, pars stress reaction, transverse process avulsion (rare)

Imaging: AP/lateral X-ray: alignment, spondylolisthesis Meyerding grade, disc height, lordosis; MRI: disc hydration (Pfarrmann), nerve root compression, Modic endplate changes, stenosis; CT: pars defects, facet tropism, surgical planning

Healing: Bone: 6-12 weeks uncomplicated fracture; pars non-union possible in spondylolysis; disc: limited intrinsic healing; annular repair fibrotic; modifiable via load management and rehab

Rehabilitation: Direction preference (flexion vs extension) per assessment; core stabilization; progressive loading; avoid prolonged bed rest; motor control multifidus/transversus; return to activity when pain controlled and functional goals met

References: Moore Clinically Oriented Anatomy; Bogduk Clinical Anatomy of the Lumbar Spine and Sacrum 2012; Netter Atlas; Gray's Anatomy

RECORD: L5 Vertebra

ID: VERT-L-05

Vertebral Level: L5

Name: Fifth lumbar vertebra

Region: Lumbar spine — lumbosacral transition

Bone Type: Irregular vertebra (lumbar type)

Body: Largest and most massive lumbar body; wedge-shaped tendency (anterior taller); sacralization/common transitional anatomy possible; bears greatest lumbosacral shear

Pedicles: L5 pedicles short and strong; project posterolaterally from body; form lateral vertebral canal boundary; pedicle screw trajectory target in surgery; nerve root exits below pedicle at same numbered level (e.g., L4 root under L4 pedicle)

Lamina: Broad laminae unite spinous process to articular pillars; ligamentum flavum attaches to anterior surface; multifidus origin from lamina and spinous process; laminectomy removes lamina for decompression

Spinous Process: Broad quadrilateral spinous process; horizontal orientation (vs thoracic oblique); supraspinous and interspinous ligament attachment; palpable midline landmark; L5 SP often prominent at lumbosacral junction

Transverse Processes: Thickest and shortest TP relative to body; robust iliolumbar ligament attachment; highest stress at lumbosacral junction

Pars Interarticularis: Pars interarticularis (pedicle-lamina junction) between superior and inferior articular processes; stress concentration site for spondylolysis; L5 pars most commonly affected in athletes; orientation permits sagittal plane motion at inferior facets; L5 pars defects common in spondylolysis/spondylolisthesis (isthmic); Scottie dog sign on oblique radiograph

Articular Facets: Superior articular facets face medially-posteriorly; inferior facets face laterally-anteriorly; paired zygapophyseal joints with L4 superiorly and L5 inferiorly; hyaline cartilage; synovial joint

Facet Orientation: ~45-55 deg (most sagittal lumbar level; transitional to sacrum)

Vertebral Canal: Triangular canal containing cauda equina (below ~L1-L2 conus); dura mater, arachnoid, CSF; epidural space; ligamentum flavum forms posterior wall; stenosis reduces cross-sectional area

Intervertebral Foramina: Formed by pedicles superior/inferior, vertebral body/disc anteriorly, facet joints posteriorly; L5 nerve root exits foramen; far-lateral/foraminal disc herniation can compress exiting root; foraminal height decreases in extension and disc height loss

Articulations: L5-L51 disc (symphysis); bilateral facet joints; L5-S1 lumbosacral junction — highest mobility and load

Muscle Attachments: Multifidus (SP, lamina, mamillary); erector spinae — iliocostalis lumborum, longissimus thoracis (TP, angles); quadratus lumborum (TP L1-L4, iliolumbar); psoas major (bodies/Tps L1-L5); latissimus dorsi via thoracolumbar fascia; intertransversarii and rotatores segmental

Ligament Attachments: ALL on anterior body; PLL posterior body; ligamentum flavum lamina-lamina; interspinous and supraspinous between SPs; intertransverse between TPs; iliolumbar ligament L5 TP to iliac crest (major stabilizer)

Blood Supply: Segmental lumbar arteries from aorta (typically 4 pairs); spinal branches through intervertebral foramen; metameric supply to body, pedicle, lamina; L5 radicular artery may contribute (variable; artery of Adamkiewicz usually T9-L2)

Venous Drainage: Segmental lumbar veins to inferior vena cava/ascending lumbar vein; internal vertebral venous plexus (Batson); valveless plexus permits metastatic spread; epidural venous engorgement in stenosis

Innervation: Segmental L5 medial branch dorsal rami innervate facet capsules and multifidus; L5 ventral ramus contributes to lumbar plexus branches; recurrent sinuvertebral nerve to disc/PLL

Biomechanical Role: L5 contributes to lordosis (~40-60 deg total lumbar); primary lumbosacral load transfer; resists anterior shear of L5 on sacrum

Normal Motion: Segmental flexion 8-12 deg, extension 4-6 deg, lateral flexion 4-6 deg, rotation 1-2 deg (rotation minimal in lumbar); coupled motions; L4-L5 and L5-S1 greatest total ROM

Load Transmission: 80% axial load through anterior column (body/disc); 20% through facet joints at neutral standing (approximate); facets share more load in extension; disc bears compression; posterior elements resist shear

Clinical Importance: L5 common site for degenerative change, facet arthropathy, and disc pathology; L5 pars/lsthesis screening in adolescent athletes

Common Injuries: Disc herniation, facet joint sprain/arthropathy, compression fracture (osteoporosis), spondylolysis (L5>L4), spondylolisthesis, pars stress reaction, transverse process avulsion (rare)

Imaging: AP/lateral X-ray: alignment, spondylolisthesis Meyerding grade, disc height, lordosis; MRI: disc hydration (Pfirrmann), nerve root compression, Modic endplate changes, stenosis; CT: pars defects, facet tropism, surgical planning

Healing: Bone: 6-12 weeks uncomplicated fracture; pars non-union possible in spondylolysis; disc: limited intrinsic healing; annular repair fibrotic; modifiable via load management and rehab

Rehabilitation: Direction preference (flexion vs extension) per assessment; core stabilization; progressive loading; avoid prolonged bed rest; motor control multifidus/transversus; return to activity when pain controlled and functional goals met

References: Moore Clinically Oriented Anatomy; Bogduk Clinical Anatomy of the Lumbar Spine and Sacrum 2012; Netter Atlas; Gray's Anatomy

2. Intervertebral Discs

RECORD: Disc L1-L2

ID: DISC-L1L2

Level: L1-L2

Annulus Fibrosus: Concentric lamellae of type I collagen (outer) and type II (inner); oblique fiber orientation (~30 deg); posterior annulus thinner than anterior; outer third innervated by sinuvertebral nerve; Sharpey fibers anchor to endplate and ring apophysis; circumferential tears (annular fissures) with degeneration

Nucleus Pulposus: Gelatinous proteoglycan-rich core (aggrecan, water ~70-90% young adult); notochordal remnant; imbibition-dehydration cycle with loading; migrates posteriorly with flexion loading (clinical relevance for flexion-intolerant LBP)

Endplates: Hyaline cartilage (~0.6-1 mm) bonded to vertebral body; semipermeable for disc nutrition diffusion; Modic type 1 (edema/inflammation), type 2 (fatty), type 3 (sclerosis) on MRI indicate endplate-bone marrow changes

Hydration: Highest water content in young adults; decreases with age; T2 MRI signal reflects hydration (Pfirrmann grade I-V); L1-L2 typically less symptomatic than lower lumbar

Blood Supply: Avascular in adult nucleus and inner annulus after ~ age 20; nutrition via diffusion through endplates and annulus periphery; limited healing capacity after annular tear

Nutrition: Mechanical pumping with diurnal loading; impaired nutrition with endplate sclerosis, smoking, sustained static loading; concentration gradients for glucose and oxygen via endplate

Innervation: Sinuvertebral (recurrent meningeal) nerve: posterior annulus, PLL; outer annulus only; nucleus aneural; pain from annular tear, endplate injury, inflammatory mediators (PLIF, TNF-alpha in disc pathology literature)

Biomechanics: Viscoelastic behavior; diurnal height loss ~1% with upright activity; intradiscal pressure highest sitting (~0.27 MPa approx standing, higher slouched sitting per Nachemson classic studies); flexion increases posterior annular stress; compression resists axial load

Compression: Primary load mode standing/lifting; nucleus pressurizes, annulus tensile hoop stress; vertical disc height 8-15 mm lumbar (level dependent); endplate failure before annulus in some compression injuries

Shear: Resisted by facet joints and posterior annulus; lumbosacral junction highest anterior shear (L5 on S1); iliolumbar ligament and facets resist shear; relevant in spondylolisthesis mechanics

Torsion: Limited lumbar rotation (~1-2 deg/segment); facets and fibers resist torsion; combined flexion-rotation increases herniation risk in biomechanical models

Degeneration: Pfirrmann grading I (homogeneous bright) to V (collapsed dark); dehydration, fissures, height loss, osteophytes; common imaging finding; weak correlation with pain alone — clinical correlation required

Bulging: Circumferential or broad-based annular extension beyond disc margin (>50% circumference or 3 mm beyond); often age-related; may indent thecal sac without focal nerve displacement

Protrusion: Focal base broader than dome; annulus outer fibers intact; contained herniation; may contact nerve root in lateral recess or foramen depending on location (central, paracentral, foraminal, extraforaminal)

Extrusion: Annular defect; herniated material base narrower than apex; may migrate craniocaudally; often symptomatic if compresses nerve root or causes inflammation

Sequestration: Free fragment separated from parent disc; may migrate; surgical consultation often considered if progressive deficit; MRI best demonstrates migrated fragment

Clinical Presentation: L1-L2 radiculopathy if nerve compressed (dermatome/myotome per level); central herniation: back pain ± bilateral symptoms; large central: cauda equina risk (EMERGENCY); leg pain worse than back pain typical radicular pattern; Valsalva may aggravate

MRI Findings: T2 signal disc hydration; HIZ (high-intensity zone) annular tear; nerve root displacement/compression; Modic endplate changes; differentiate bulge vs protrusion vs extrusion vs sequestration per standardized nomenclature

Treatment: Conservative first for uncomplicated radiculopathy: education, stay active, NSAIDs, PT; epidural steroid injection selective cases; surgery (microdiscectomy) for persistent radiculopathy or cauda equina; NICE NG59 / APTA LBP CPG: avoid bed rest, reassurance, graded activity

Rehabilitation: McKenzie directional preference if applicable; neural mobilization; core stabilization; progressive walking and loading; avoid prolonged flexion early if flexion-intolerant; return to sport when pain controlled, strength symmetric, no neurological progression

References: Bogduk 2012; Jensen et al. N Engl J Med disc herniation imaging; Pfirrmann classification; NICE NG59; APTA Low Back Pain CPG

RECORD: Disc L2-L3

ID: DISC-L2L3

Level: L2-L3

Annulus Fibrosus: Concentric lamellae of type I collagen (outer) and type II (inner); oblique fiber orientation (~30 deg); posterior annulus thinner than anterior; outer third innervated by sinuvertebral nerve; Sharpey fibers anchor to endplate and ring apophysis; circumferential tears (annular fissures) with degeneration

Nucleus Pulposus: Gelatinous proteoglycan-rich core (aggrecan, water ~70-90% young adult); notochordal remnant; imbibition-dehydration cycle with loading; migrates posteriorly with flexion loading (clinical relevance for flexion-intolerant LBP)

Endplates: Hyaline cartilage (~0.6-1 mm) bonded to vertebral body; semipermeable for disc nutrition diffusion; Modic type 1 (edema/inflammation), type 2 (fatty), type 3 (sclerosis) on MRI indicate endplate-bone marrow changes

Hydration: Highest water content in young adults; decreases with age; T2 MRI signal reflects hydration (Pfirrmann grade I-V); L2-L3 typically less symptomatic than lower lumbar

Blood Supply: Avascular in adult nucleus and inner annulus after ~ age 20; nutrition via diffusion through endplates and annulus periphery; limited healing capacity after annular tear

Nutrition: Mechanical pumping with diurnal loading; impaired nutrition with endplate sclerosis, smoking, sustained static loading; concentration gradients for glucose and oxygen via endplate

Innervation: Sinuvertebral (recurrent meningeal) nerve: posterior annulus, PLL; outer annulus only; nucleus aneural; pain from annular tear, endplate injury, inflammatory mediators (PLIF, TNF-alpha in disc pathology literature)

Biomechanics: Viscoelastic behavior; diurnal height loss ~1% with upright activity; intradiscal pressure highest sitting (~0.27 MPa approx standing, higher slouched sitting per Nachemson classic studies); flexion increases posterior annular stress; compression resists axial load

Compression: Primary load mode standing/lifting; nucleus pressurizes, annulus tensile hoop stress; vertical disc height 8-15 mm lumbar (level dependent); endplate failure before annulus in some compression injuries

Shear: Resisted by facet joints and posterior annulus; lumbosacral junction highest anterior shear (L5 on S1); iliolumbar ligament and facets resist shear; relevant in spondylolisthesis mechanics

Torsion: Limited lumbar rotation (~1-2 deg/segment); facets and fibers resist torsion; combined flexion-rotation increases herniation risk in biomechanical models

Degeneration: Pfirrmann grading I (homogeneous bright) to V (collapsed dark); dehydration, fissures, height loss, osteophytes; common imaging finding; weak correlation with pain alone — clinical correlation required

Bulging: Circumferential or broad-based annular extension beyond disc margin (>50% circumference or 3 mm beyond); often age-related; may indent thecal sac without focal nerve displacement

Protrusion: Focal base broader than dome; annulus outer fibers intact; contained herniation; may contact nerve root in lateral recess or foramen depending on location (central, paracentral, foraminal, extraforaminal)

Extrusion: Annular defect; herniated material base narrower than apex; may migrate craniocaudally; often symptomatic if compresses nerve root or causes inflammation

Sequestration: Free fragment separated from parent disc; may migrate; surgical consultation often considered if progressive deficit; MRI best demonstrates migrated fragment

Clinical Presentation: L2-L3 radiculopathy if nerve compressed (dermatome/myotome per level); central herniation: back pain ± bilateral symptoms; large central: cauda equina risk (EMERGENCY); leg pain worse than back pain typical radicular pattern; Valsalva may aggravate

MRI Findings: T2 signal disc hydration; HIZ (high-intensity zone) annular tear; nerve root displacement/compression; Modic endplate changes; differentiate bulge vs protrusion vs extrusion vs sequestration per standardized nomenclature

Treatment: Conservative first for uncomplicated radiculopathy: education, stay active, NSAIDs, PT; epidural steroid injection selective cases; surgery (microdiscectomy) for persistent radiculopathy or cauda equina; NICE NG59 / APTA LBP CPG: avoid bed rest, reassurance, graded activity

Rehabilitation: McKenzie directional preference if applicable; neural mobilization; core stabilization; progressive walking and loading; avoid prolonged flexion early if flexion-intolerant; return to sport when pain controlled, strength symmetric, no neurological progression

References: Bogduk 2012; Jensen et al. N Engl J Med disc herniation imaging; Pfirrmann classification; NICE NG59; APTA Low Back Pain CPG

RECORD: Disc L3-L4

ID: DISC-L3L4

Level: L3-L4

Annulus Fibrosus: Concentric lamellae of type I collagen (outer) and type II (inner); oblique fiber orientation (~30 deg); posterior annulus thinner than anterior; outer third innervated by sinuvertebral nerve; Sharpey fibers anchor to endplate and ring apophysis; circumferential tears (annular fissures) with degeneration

Nucleus Pulposus: Gelatinous proteoglycan-rich core (aggrecan, water ~70-90% young adult); notochordal remnant; imbibition-dehydration cycle with loading; migrates posteriorly with flexion loading (clinical relevance for flexion-intolerant LBP)

Endplates: Hyaline cartilage (~0.6-1 mm) bonded to vertebral body; semipermeable for disc nutrition diffusion; Modic type 1 (edema/inflammation), type 2 (fatty), type 3 (sclerosis) on MRI indicate endplate-bone marrow changes

Hydration: Highest water content in young adults; decreases with age; T2 MRI signal reflects hydration (Pfirrmann grade I-V); L3-L4 typically less symptomatic than lower lumbar

Blood Supply: Avascular in adult nucleus and inner annulus after ~ age 20; nutrition via diffusion through endplates and annulus periphery; limited healing capacity after annular tear

Nutrition: Mechanical pumping with diurnal loading; impaired nutrition with endplate sclerosis, smoking, sustained static loading; concentration gradients for glucose and oxygen via endplate

Innervation: Sinuvertebral (recurrent meningeal) nerve: posterior annulus, PLL; outer annulus only; nucleus aneural; pain from annular tear, endplate injury, inflammatory mediators (PLIF, TNF-alpha in disc pathology literature)

Biomechanics: Viscoelastic behavior; diurnal height loss ~1% with upright activity; intradiscal pressure highest sitting (~0.27 MPa approx standing, higher slouched sitting per Nachemson classic studies); flexion increases posterior annular stress; compression resists axial load

Compression: Primary load mode standing/lifting; nucleus pressurizes, annulus tensile hoop stress; vertical disc height 8-15 mm lumbar (level dependent); endplate failure before annulus in some compression injuries

Shear: Resisted by facet joints and posterior annulus; lumbosacral junction highest anterior shear (L5 on S1); iliolumbar ligament and facets resist shear; relevant in spondylolisthesis mechanics

Torsion: Limited lumbar rotation (~1-2 deg/segment); facets and fibers resist torsion; combined flexion-rotation increases herniation risk in biomechanical models

Degeneration: Pfirrmann grading I (homogeneous bright) to V (collapsed dark); dehydration, fissures, height loss, osteophytes; common imaging finding; weak correlation with pain alone — clinical correlation required

Bulging: Circumferential or broad-based annular extension beyond disc margin (>50% circumference or 3 mm beyond); often age-related; may indent thecal sac without focal nerve displacement

Protrusion: Focal base broader than dome; annulus outer fibers intact; contained herniation; may contact nerve root in lateral recess or foramen depending on location (central, paracentral, foraminal, extraforaminal)

Extrusion: Annular defect; herniated material base narrower than apex; may migrate craniocaudally; often symptomatic if compresses nerve root or causes inflammation

Sequestration: Free fragment separated from parent disc; may migrate; surgical consultation often considered if progressive deficit; MRI best demonstrates migrated fragment

Clinical Presentation: L3-L4 radiculopathy if nerve compressed (dermatome/myotome per level); central herniation: back pain ± bilateral symptoms; large central: cauda equina risk (EMERGENCY); leg pain worse than back pain typical radicular

pattern; Valsalva may aggravate

MRI Findings: T2 signal disc hydration; HIZ (high-intensity zone) annular tear; nerve root displacement/compression; Modic endplate changes; differentiate bulge vs protrusion vs extrusion vs sequestration per standardized nomenclature

Treatment: Conservative first for uncomplicated radiculopathy: education, stay active, NSAIDs, PT; epidural steroid injection selective cases; surgery (microdiscectomy) for persistent radiculopathy or cauda equina; NICE NG59 / APTA LBP CPG: avoid bed rest, reassurance, graded activity

Rehabilitation: McKenzie directional preference if applicable; neural mobilization; core stabilization; progressive walking and loading; avoid prolonged flexion early if flexion-intolerant; return to sport when pain controlled, strength symmetric, no neurological progression

References: Bogduk 2012; Jensen et al. N Engl J Med disc herniation imaging; Pfirrmann classification; NICE NG59; APTA Low Back Pain CPG

RECORD: Disc L4-L5

ID: DISC-L4L5

Level: L4-L5

Annulus Fibrosus: Concentric lamellae of type I collagen (outer) and type II (inner); oblique fiber orientation (~30 deg); posterior annulus thinner than anterior; outer third innervated by sinuvertebral nerve; Sharpey fibers anchor to endplate and ring apophysis; circumferential tears (annular fissures) with degeneration

Nucleus Pulposus: Gelatinous proteoglycan-rich core (aggrecan, water ~70-90% young adult); notochordal remnant; imbibition-dehydration cycle with loading; migrates posteriorly with flexion loading (clinical relevance for flexion-intolerant LBP)

Endplates: Hyaline cartilage (~0.6-1 mm) bonded to vertebral body; semipermeable for disc nutrition diffusion; Modic type 1 (edema/inflammation), type 2 (fatty), type 3 (sclerosis) on MRI indicate endplate-bone marrow changes

Hydration: Highest water content in young adults; decreases with age; T2 MRI signal reflects hydration (Pfirrmann grade I-V); L4-L5 second most common symptomatic level

Blood Supply: Avascular in adult nucleus and inner annulus after ~ age 20; nutrition via diffusion through endplates and annulus periphery; limited healing capacity after annular tear

Nutrition: Mechanical pumping with diurnal loading; impaired nutrition with endplate sclerosis, smoking, sustained static loading; concentration gradients for glucose and oxygen via endplate

Innervation: Sinuvertebral (recurrent meningeal) nerve: posterior annulus, PLL; outer annulus only; nucleus aneural; pain from annular tear, endplate injury, inflammatory mediators (PLIF, TNF-alpha in disc pathology literature)

Biomechanics: Viscoelastic behavior; diurnal height loss ~1% with upright activity; intradiscal pressure highest sitting (~0.27 MPa approx standing, higher slouched sitting per Nachemson classic studies); flexion increases posterior annular stress; compression resists axial load

Compression: Primary load mode standing/lifting; nucleus pressurizes, annulus tensile hoop stress; vertical disc height 8-15 mm lumbar (level dependent); endplate failure before annulus in some compression injuries

Shear: Resisted by facet joints and posterior annulus; lumbosacral junction highest anterior shear (L5 on S1); iliolumbar ligament and facets resist shear; relevant in spondylolisthesis mechanics

Torsion: Limited lumbar rotation (~1-2 deg/segment); facets and fibers resist torsion; combined flexion-rotation increases herniation risk in biomechanical models

Degeneration: Pfirrmann grading I (homogeneous bright) to V (collapsed dark); dehydration, fissures, height loss, osteophytes; common imaging finding; weak correlation with pain alone — clinical correlation required

Bulging: Circumferential or broad-based annular extension beyond disc margin (>50% circumference or 3 mm beyond); often age-related; may indent thecal sac without focal nerve displacement

Protrusion: Focal base broader than dome; annulus outer fibers intact; contained herniation; may contact nerve root in lateral recess or foramen depending on location (central, paracentral, foraminal, extraforaminal)

Extrusion: Annular defect; herniated material base narrower than apex; may migrate craniocaudally; often symptomatic if compresses nerve root or causes inflammation

Sequestration: Free fragment separated from parent disc; may migrate; surgical consultation often considered if progressive deficit; MRI best demonstrates migrated fragment

Clinical Presentation: L4-L5 radiculopathy if nerve compressed (dermatome/myotome per level); central herniation: back pain ± bilateral symptoms; large central: cauda equina risk (EMERGENCY); leg pain worse than back pain typical radicular pattern; Valsalva may aggravate

MRI Findings: T2 signal disc hydration; HIZ (high-intensity zone) annular tear; nerve root displacement/compression; Modic endplate changes; differentiate bulge vs protrusion vs extrusion vs sequestration per standardized nomenclature

Treatment: Conservative first for uncomplicated radiculopathy: education, stay active, NSAIDs, PT; epidural steroid injection selective cases; surgery (microdiscectomy) for persistent radiculopathy or cauda equina; NICE NG59 / APTA LBP CPG: avoid bed rest, reassurance, graded activity

Rehabilitation: McKenzie directional preference if applicable; neural mobilization; core stabilization; progressive walking and loading; avoid prolonged flexion early if flexion-intolerant; return to sport when pain controlled, strength symmetric, no neurological progression

References: Bogduk 2012; Jensen et al. N Engl J Med disc herniation imaging; Pfirrmann classification; NICE NG59; APTA Low Back Pain CPG

RECORD: Disc L5-S1

ID: DISC-L5S1

Level: L5-S1

Annulus Fibrosus: Concentric lamellae of type I collagen (outer) and type II (inner); oblique fiber orientation (~30 deg); posterior annulus thinner than anterior; outer third innervated by sinuvertebral nerve; Sharpey fibers anchor to endplate and ring apophysis; circumferential tears (annular fissures) with degeneration

Nucleus Pulposus: Gelatinous proteoglycan-rich core (aggrecan, water ~70-90% young adult); notochordal remnant; imbibition-dehydration cycle with loading; migrates posteriorly with flexion loading (clinical relevance for flexion-intolerant LBP)

Endplates: Hyaline cartilage (~0.6-1 mm) bonded to vertebral body; semipermeable for disc nutrition diffusion; Modic type 1 (edema/inflammation), type 2 (fatty), type 3 (sclerosis) on MRI indicate endplate-bone marrow changes

Hydration: Highest water content in young adults; decreases with age; T2 MRI signal reflects hydration (Pfirrmann grade I-V); L5-S1 earliest/most common degeneration

Blood Supply: Avascular in adult nucleus and inner annulus after ~ age 20; nutrition via diffusion through endplates and annulus periphery; limited healing capacity after annular tear

Nutrition: Mechanical pumping with diurnal loading; impaired nutrition with endplate sclerosis, smoking, sustained static loading; concentration gradients for glucose and oxygen via endplate

Innervation: Sinuvertebral (recurrent meningeal) nerve: posterior annulus, PLL; outer annulus only; nucleus aneural; pain from annular tear, endplate injury, inflammatory mediators (PLIF, TNF-alpha in disc pathology literature)

Biomechanics: Viscoelastic behavior; diurnal height loss ~1% with upright activity; intradiscal pressure highest sitting (~0.27 MPa approx standing, higher slouched sitting per Nachemson classic studies); flexion increases posterior annular stress; compression resists axial load

Compression: Primary load mode standing/lifting; nucleus pressurizes, annulus tensile hoop stress; vertical disc height 8-15 mm lumbar (level dependent); endplate failure before annulus in some compression injuries

Shear: Resisted by facet joints and posterior annulus; lumbosacral junction highest anterior shear (L5 on S1); iliolumbar ligament and facets resist shear; relevant in spondylolisthesis mechanics

Torsion: Limited lumbar rotation (~1-2 deg/segment); facets and fibers resist torsion; combined flexion-rotation increases herniation risk in biomechanical models

Degeneration: Pfirrmann grading I (homogeneous bright) to V (collapsed dark); dehydration, fissures, height loss, osteophytes; common imaging finding; weak correlation with pain alone — clinical correlation required

Bulging: Circumferential or broad-based annular extension beyond disc margin (>50% circumference or 3 mm beyond); often age-related; may indent thecal sac without focal nerve displacement

Protrusion: Focal base broader than dome; annulus outer fibers intact; contained herniation; may contact nerve root in lateral recess or foramen depending on location (central, paracentral, foraminal, extraforaminal)

Extrusion: Annular defect; herniated material base narrower than apex; may migrate craniocaudally; often symptomatic if compresses nerve root or causes inflammation

Sequestration: Free fragment separated from parent disc; may migrate; surgical consultation often considered if progressive deficit; MRI best demonstrates migrated fragment

Clinical Presentation: L5-S1 radiculopathy if nerve compressed (dermatome/myotome per level); central herniation: back pain ± bilateral symptoms; large central: cauda equina risk (EMERGENCY); leg pain worse than back pain typical radicular pattern; Valsalva may aggravate

MRI Findings: T2 signal disc hydration; HIZ (high-intensity zone) annular tear; nerve root displacement/compression; Modic endplate changes; differentiate bulge vs protrusion vs extrusion vs sequestration per standardized nomenclature

Treatment: Conservative first for uncomplicated radiculopathy: education, stay active, NSAIDs, PT; epidural steroid injection selective cases; surgery (microdiscectomy) for persistent radiculopathy or cauda equina; NICE NG59 / APTA LBP CPG: avoid bed rest, reassurance, graded activity

Rehabilitation: McKenzie directional preference if applicable; neural mobilization; core stabilization; progressive walking and loading; avoid prolonged flexion early if flexion-intolerant; return to sport when pain controlled, strength symmetric, no neurological progression

References: Bogduk 2012; Jensen et al. N Engl J Med disc herniation imaging; Pfirrmann classification; NICE NG59; APTA Low Back Pain CPG

3. Facet Joints

RECORD: Facet Joint L1-L2

ID: FACET-L1L2

Level: L1-L2

Orientation: Biomechanical plane ~90 deg to sagittal (coronal dominant); resists rotation, permits limited flexion-extension

Joint Type: Synovial zygapophyseal (plane) joint; hyaline cartilage; fibrous capsule

Capsule: Loose fibrous capsule; innervated by medial branch dorsal ramus; capsular distension/synovitis source of facet-mediated pain; fibroadipose meniscoids at superior-inferior poles

Ligaments: Facet capsule reinforced by adjacent ligamentum flavum (posterior); ALL/PLL indirect influence; no separate named ligament beyond capsule at lumbar facets

Blood Supply: Medial branch of dorsal ramus accompanying artery; capillary supply to capsule and articular cartilage (limited)

Innervation: Medial branch dorsal ramus of same level and level below (dual innervation pattern); facet denervation via medial branch block/radiofrequency ablation targets in chronic facet pain workup

Arthrokinematics: Extension: inferior facet slides inferior-posterior on superior facet of vertebra below; flexion: superior glide; gapping on ipsilateral side bending; close-pack position extension; open-pack flexion-rotation

Osteokinematics: Guide and limit lumbar motion; sagittal plane primary; rotation minimal; facet asymmetry (tropism) alters motion and may predispose to degeneration

Normal Motion: Segmental contribution to total lumbar ROM; L1-L2 flexion-extension primary; rotation restricted by facet orientation and disc

Pain Referral: Deep axial low back pain; may refer to buttock, posterior thigh (sclerotomal pattern); typically NOT below knee (vs radicular); paraspinal tenderness 1-2 cm lateral to SP

Facet Syndrome: Clinical diagnosis of facet-mediated pain; extension/rotation loading reproduces pain; relief with flexion; common in extension-intolerant LBP; degenerative facet arthropathy on imaging common with age

Special Tests: Extension-rotation test; Kemp's quadrant (extension-rotation compression); facet loading in standing extension; PA spring test over articular pillar; diagnostic medial branch block (≥80% relief threshold in research protocols for facet pain confirmation)

Manual Therapy: Grade III-IV PA mobilization; flexion distraction techniques; HVLA manipulation when appropriate and no contraindications; avoid aggressive extension manipulation in stenosis

Rehabilitation: Flexion-biased exercises if extension provocative; core stabilization; hip hinge pattern training; avoid prolonged lumbar extension loading; aerobic conditioning; motor control for neutral spine with functional tasks

References: Bogduk 2012 facet joint innervation; Magee 2014; Cohen et al. facet joint interventions systematic reviews

RECORD: Facet Joint L2-L3

ID: FACET-L2L3

Level: L2-L3

Orientation: ~85 deg coronal; increasing sagittal component inferiorly

Joint Type: Synovial zygapophyseal (plane) joint; hyaline cartilage; fibrous capsule

Capsule: Loose fibrous capsule; innervated by medial branch dorsal ramus; capsular distension/synovitis source of facet-mediated pain; fibroadipose meniscoids at superior-inferior poles

Ligaments: Facet capsule reinforced by adjacent ligamentum flavum (posterior); ALL/PLL indirect influence; no separate named ligament beyond capsule at lumbar facets

Blood Supply: Medial branch of dorsal ramus accompanying artery; capillary supply to capsule and articular cartilage (limited)

Innervation: Medial branch dorsal ramus of same level and level below (dual innervation pattern); facet denervation via medial branch block/radiofrequency ablation targets in chronic facet pain workup

Arthrokinematics: Extension: inferior facet slides inferior-posterior on superior facet of vertebra below; flexion: superior glide; gapping on ipsilateral side bending; close-pack position extension; open-pack flexion-rotation

Osteokinematics: Guide and limit lumbar motion; sagittal plane primary; rotation minimal; facet asymmetry (tropism) alters motion and may predispose to degeneration

Normal Motion: Segmental contribution to total lumbar ROM; L2-L3 flexion-extension primary; rotation restricted by facet orientation and disc

Pain Referral: Deep axial low back pain; may refer to buttock, posterior thigh (sclerotomal pattern); typically NOT below knee (vs radicular); paraspinal tenderness 1-2 cm lateral to SP

Facet Syndrome: Clinical diagnosis of facet-mediated pain; extension/rotation loading reproduces pain; relief with flexion; common in extension-intolerant LBP; degenerative facet arthropathy on imaging common with age

Special Tests: Extension-rotation test; Kemp's quadrant (extension-rotation compression); facet loading in standing extension; PA spring test over articular pillar; diagnostic medial branch block ($\geq 80\%$ relief threshold in research protocols for facet pain confirmation)

Manual Therapy: Grade III-IV PA mobilization; flexion distraction techniques; HVLA manipulation when appropriate and no contraindications; avoid aggressive extension manipulation in stenosis

Rehabilitation: Flexion-biased exercises if extension provocative; core stabilization; hip hinge pattern training; avoid prolonged lumbar extension loading; aerobic conditioning; motor control for neutral spine with functional tasks

References: Bogduk 2012 facet joint innervation; Magee 2014; Cohen et al. facet joint interventions systematic reviews

RECORD: Facet Joint L3-L4

ID: FACET-L3L4

Level: L3-L4

Orientation: ~75 deg; transitional orientation

Joint Type: Synovial zygapophyseal (plane) joint; hyaline cartilage; fibrous capsule

Capsule: Loose fibrous capsule; innervated by medial branch dorsal ramus; capsular distension/synovitis source of facet-mediated pain; fibroadipose meniscoids at superior-inferior poles

Ligaments: Facet capsule reinforced by adjacent ligamentum flavum (posterior); ALL/PLL indirect influence; no separate named ligament beyond capsule at lumbar facets

Blood Supply: Medial branch of dorsal ramus accompanying artery; capillary supply to capsule and articular cartilage (limited)

Innervation: Medial branch dorsal ramus of same level and level below (dual innervation pattern); facet denervation via medial branch block/radiofrequency ablation targets in chronic facet pain workup

Arthrokinematics: Extension: inferior facet slides inferior-posterior on superior facet of vertebra below; flexion: superior glide; gapping on ipsilateral side bending; close-pack position extension; open-pack flexion-rotation

Osteokinematics: Guide and limit lumbar motion; sagittal plane primary; rotation minimal; facet asymmetry (tropism) alters motion and may predispose to degeneration

Normal Motion: Segmental contribution to total lumbar ROM; L3-L4 flexion-extension primary; rotation restricted by facet orientation and disc

Pain Referral: Deep axial low back pain; may refer to buttock, posterior thigh (sclerotomal pattern); typically NOT below knee (vs radicular); paraspinal tenderness 1-2 cm lateral to SP

Facet Syndrome: Clinical diagnosis of facet-mediated pain; extension/rotation loading reproduces pain; relief with flexion; common in extension-intolerant LBP; degenerative facet arthropathy on imaging common with age

Special Tests: Extension-rotation test; Kemp's quadrant (extension-rotation compression); facet loading in standing extension; PA spring test over articular pillar; diagnostic medial branch block ($\geq 80\%$ relief threshold in research protocols for facet pain confirmation)

Manual Therapy: Grade III-IV PA mobilization; flexion distraction techniques; HVLA manipulation when appropriate and no contraindications; avoid aggressive extension manipulation in stenosis

Rehabilitation: Flexion-biased exercises if extension provocative; core stabilization; hip hinge pattern training; avoid prolonged lumbar extension loading; aerobic conditioning; motor control for neutral spine with functional tasks

References: Bogduk 2012 facet joint innervation; Magee 2014; Cohen et al. facet joint interventions systematic reviews

RECORD: Facet Joint L4-L5

ID: FACET-L4L5

Level: L4-L5

Orientation: ~60 deg; more sagittal — increased flexion-extension freedom

Joint Type: Synovial zygapophyseal (plane) joint; hyaline cartilage; fibrous capsule

Capsule: Loose fibrous capsule; innervated by medial branch dorsal ramus; capsular distension/synovitis source of facet-mediated pain; fibroadipose meniscoids at superior-inferior poles

Ligaments: Facet capsule reinforced by adjacent ligamentum flavum (posterior); ALL/PLL indirect influence; no separate named ligament beyond capsule at lumbar facets

Blood Supply: Medial branch of dorsal ramus accompanying artery; capillary supply to capsule and articular cartilage (limited)

Innervation: Medial branch dorsal ramus of same level and level below (dual innervation pattern); facet denervation via medial branch block/radiofrequency ablation targets in chronic facet pain workup

Arthrokinematics: Extension: inferior facet slides inferior-posterior on superior facet of vertebra below; flexion: superior glide; gapping on ipsilateral side bending; close-pack position extension; open-pack flexion-rotation

Osteokinematics: Guide and limit lumbar motion; sagittal plane primary; rotation minimal; facet asymmetry (tropism) alters motion and may predispose to degeneration

Normal Motion: Segmental contribution to total lumbar ROM; L4-L5 flexion-extension primary; rotation restricted by facet orientation and disc

Pain Referral: Deep axial low back pain; may refer to buttock, posterior thigh (sclerotomal pattern); typically NOT below knee (vs radicular); paraspinal tenderness 1-2 cm lateral to SP

Facet Syndrome: Clinical diagnosis of facet-mediated pain; extension/rotation loading reproduces pain; relief with flexion; common in extension-intolerant LBP; degenerative facet arthropathy on imaging common with age

Special Tests: Extension-rotation test; Kemp's quadrant (extension-rotation compression); facet loading in standing extension; PA spring test over articular pillar; diagnostic medial branch block (≥80% relief threshold in research protocols for facet pain confirmation)

Manual Therapy: Grade III-IV PA mobilization; flexion distraction techniques; HVLA manipulation when appropriate and no contraindications; avoid aggressive extension manipulation in stenosis

Rehabilitation: Flexion-biased exercises if extension provocative; core stabilization; hip hinge pattern training; avoid prolonged lumbar extension loading; aerobic conditioning; motor control for neutral spine with functional tasks

References: Bogduk 2012 facet joint innervation; Magee 2014; Cohen et al. facet joint interventions systematic reviews

RECORD: Facet Joint L5-S1

ID: FACET-L5S1

Level: L5-S1

Orientation: ~45-50 deg sagittal dominant at L5-S1; greatest flexion-extension, highest facet load in extension

Joint Type: Synovial zygapophyseal (plane) joint; hyaline cartilage; fibrous capsule

Capsule: Loose fibrous capsule; innervated by medial branch dorsal ramus; capsular distension/synovitis source of facet-mediated pain; fibroadipose meniscoids at superior-inferior poles

Ligaments: Facet capsule reinforced by adjacent ligamentum flavum (posterior); ALL/PLL indirect influence; no separate named ligament beyond capsule at lumbar facets

Blood Supply: Medial branch of dorsal ramus accompanying artery; capillary supply to capsule and articular cartilage (limited)

Innervation: Medial branch dorsal ramus of same level and level below (dual innervation pattern); facet denervation via medial branch block/radiofrequency ablation targets in chronic facet pain workup

Arthrokinematics: Extension: inferior facet slides inferior-posterior on superior facet of vertebra below; flexion: superior glide; gapping on ipsilateral side bending; close-pack position extension; open-pack flexion-rotation

Osteokinematics: Guide and limit lumbar motion; sagittal plane primary; rotation minimal; facet asymmetry (tropism) alters motion and may predispose to degeneration

Normal Motion: Segmental contribution to total lumbar ROM; L5-S1 flexion-extension primary; rotation restricted by facet orientation and disc

Pain Referral: Deep axial low back pain; may refer to buttock, posterior thigh (sclerotomal pattern); typically NOT below knee (vs radicular); paraspinal tenderness 1-2 cm lateral to SP

Facet Syndrome: Clinical diagnosis of facet-mediated pain; extension/rotation loading reproduces pain; relief with flexion; common in extension-intolerant LBP; degenerative facet arthropathy on imaging common with age

Special Tests: Extension-rotation test; Kemp's quadrant (extension-rotation compression); facet loading in standing extension; PA spring test over articular pillar; diagnostic medial branch block ($\geq 80\%$ relief threshold in research protocols for facet pain confirmation)

Manual Therapy: Grade III-IV PA mobilization; flexion distraction techniques; HVLA manipulation when appropriate and no contraindications; avoid aggressive extension manipulation in stenosis

Rehabilitation: Flexion-biased exercises if extension provocative; core stabilization; hip hinge pattern training; avoid prolonged lumbar extension loading; aerobic conditioning; motor control for neutral spine with functional tasks

References: Bogduk 2012 facet joint innervation; Magee 2014; Cohen et al. facet joint interventions systematic reviews

4. Ligaments

RECORD: Anterior Longitudinal Ligament (ALL)

ID: LIG-LUM-001

Name: Anterior Longitudinal Ligament (ALL)

Origin: Basilar occiput (atlanto-occipital) continuous inferiorly along anterior vertebral bodies and discs to sacrum

Insertion: Anterior annulus and vertebral body margins at each level; broadest in lumbar spine

Function: Resists hyperextension, distraction anteriorly, and some rotation; stabilizes anterior column

Biomechanics: Primary anterior tension band; limits disc bulge anteriorly; loaded in extension

Load Capacity: High tensile strength; ossification (DISH) may reduce segmental motion

Blood Supply: Periosteal and segmental branches; relatively avascular mid-substance

Healing: Slow fibrous healing; calcification possible with chronic injury or DISH

Clinical Tests: Hyperextension pain; not isolated ligament test — inferred from mechanism and imaging

Common Injuries: Avulsion rare; more commonly involved in hyperextension trauma with vertebral body fracture

MRI: Low signal T1/T2; thickening or ossification visible; DISH: flowing ossification along ALL

Ultrasound: Limited deep visualization; not primary modality for ALL

Treatment: Extension restriction if acute sprain; bracing rarely; progressive ROM when pain allows

Rehabilitation: Neutral spine training; avoid aggressive extension early; core stabilization

Return to Sport: When extension pain-free and functional loading tolerated

References: Moore; Bogduk 2012; Gray's Anatomy

RECORD: Posterior Longitudinal Ligament (PLL)

ID: LIG-LUM-002

Name: Posterior Longitudinal Ligament (PLL)

Origin: C2 body (cranial continuation) along posterior vertebral bodies inside vertebral canal

Insertion: Posterior annulus and posterior body margins; narrows at disc levels (hourglass shape)

Function: Resists posterior disc herniation centrally; limits flexion distraction posteriorly

Biomechanics: Weaker centrally than ALL; lateral disc herniations may occur where PLL is narrowest

Load Capacity: Moderate; ossification of PLL (OPLL) causes stenosis (more cervical but concept applies)

Blood Supply: Sinuvertebral artery branches

Healing: Limited; ossification in chronic conditions

Clinical Tests: No direct test; central disc herniation may compress thecal sac despite PLL

Common Injuries: Indirect — disc herniation under PLL; PLL rupture in severe trauma

MRI: Visible posterior to disc; OPLL if ossified; stenosis assessment

Ultrasound: Not applicable deep structure

Treatment: Conservative for disc; surgical if stenosis/myelopathy/cauda equina

Rehabilitation: Per underlying disc pathology; neural mobilization if radiculopathy

Return to Sport: Per disc herniation rehab protocol

References: Bogduk 2012; Netter

RECORD: Ligamentum Flavum

ID: LIG-LUM-003

Name: Ligamentum Flavum

Origin: Anterior surface of cranial lamina

Insertion: Posterior surface of caudal lamina; spans interlaminar space

Function: Elastin-rich ligament maintains upright posture; prevents buckling into canal in flexion; aids extension return

Biomechanics: Hypertrophy with age/degeneration contributes to central and lateral recess stenosis; buckling in extension

Load Capacity: Elastic recoil; 80% elastin content approx

Blood Supply: Segmental spinal branches

Healing: Hypertrophic fibrosis rather than normal restoration — may worsen stenosis

Clinical Tests: Stenosis symptoms worse extension-standing walking (shopping cart sign); not isolated flavum test

Common Injuries: Hypertrophy/degeneration; calcification rare

MRI: Best modality; thickening measured on axial T2; contributes to stenosis grade

Ultrasound: Not used

Treatment: Surgical decompression (laminectomy/laminotomy) when neurogenic claudication refractory

Rehabilitation: Flexion-biased mobility; stationary bike; strengthen hip flexors for flexed posture relief

Return to Sport: Walking/cycling first; avoid extension-loaded sports until symptoms controlled

References: Bogduk 2012; Magee 2014 lumbar stenosis

RECORD: Interspinous Ligament

ID: LIG-LUM-004

Name: Interspinous Ligament

Origin: Base of spinous process

Insertion: Apex of spinous process below

Function: Resists flexion separation of SPs; connects posterior column

Biomechanics: Loaded in flexion; weak link compared to supraspinous in some bending models

Load Capacity: Moderate tensile

Blood Supply: Dorsal rami branches

Healing: Fibrous healing 6-12 weeks typical sprain

Clinical Tests: Palpation midline tenderness; flexion pain

Common Injuries: Sprain with hyperflexion; may accompany extension injury of anterior structures

MRI: Edema between SPs if acute tear

Ultrasound: Limited utility

Treatment: Relative rest, NSAIDs, gradual ROM

Rehabilitation: Cat-camel, bird dog, progressive extension control

Return to Sport: Pain-free flexion-extension and rotation

References: Moore; Gray's Anatomy

RECORD: Supraspinous Ligament

ID: LIG-LUM-005

Name: Supraspinous Ligament

Origin: Tips of spinous processes

Insertion: Continuous superiorly to thoracolumbar region; merges with thoracic supraspinous

Function: Posterior tension band limiting flexion; palpable landmark

Biomechanics: Works with thoracolumbar fascia and erector spinae; loaded in forward bending

Load Capacity: Moderate; can sprain with sudden flexion load

Blood Supply: Segmental cutaneous and dorsal branches

Healing: 6-12 weeks sprain
Clinical Tests: Midline SP tenderness; forward bend pain
Common Injuries: Sprain; avulsion rare
MRI: Signal change at SP tips if torn
Ultrasound: Superficial portions occasionally imaged
Treatment: Activity modification, heat, progressive loading
Rehabilitation: Prone extension progressions if tolerated; core stability
Return to Sport: Full pain-free ROM and lifting
References: Moore; Magee 2014

RECORD: Intertransverse Ligament

ID: LIG-LUM-006
Name: Intertransverse Ligament
Origin: Transverse process
Insertion: Adjacent transverse process
Function: Limits lateral flexion; attachment site for psoas and quadratus lumborum
Biomechanics: Lateral bending restraint; weak compared to iliolumbar at L4-L5
Load Capacity: Low-moderate
Blood Supply: Segmental arteries
Healing: Standard soft tissue healing
Clinical Tests: Side bending pain; palpation lateral to SP
Common Injuries: Strain with lateral bending trauma
MRI: Uncommon isolated finding
Ultrasound: Limited
Treatment: Conservative
Rehabilitation: Lateral core stability; QL stretching/strengthening
Return to Sport: Symmetric lateral flexion strength
References: Bogduk 2012

RECORD: Iliolumbar Ligament

ID: LIG-LUM-007
Name: Iliolumbar Ligament
Origin: Transverse processes L4 and L5 (strongest L5 attachment)
Insertion: Iliac crest posteriorly
Function: Critical lumbosacral stabilizer; resists L5 anterior shear and side bending
Biomechanics: Major restraint at lumbosacral junction; stressed in lifting and rotation; weaker in spondylolisthesis
Load Capacity: High — strongest lumbar ligament by cross-sectional area
Blood Supply: Iliolumbar artery branch
Healing: Slow; chronic insufficiency contributes to instability sensation
Clinical Tests: Pelvic distortion test; palpation TP to iliac crest tenderness; not highly specific
Common Injuries: Sprain with torsion/lifting; associated with LBP and SI region pain
MRI: Difficult to visualize; edema at TP attachment if acute
Ultrasound: Limited deep structure
Treatment: Stabilization exercises; avoid heavy rotation loading acute phase

Rehabilitation: QL and multifidus; hip hinge; progressive unilateral loading

Return to Sport: Single-leg stability tests passed; no shear-provocative pain

References: Bogduk 2012; lumbosacral biomechanics texts

RECORD: Capsular Facet Ligaments

ID: LIG-LUM-008

Name: Capsular Facet Ligaments (Zygapophyseal Joint Capsule)

Origin: Margins of superior and inferior articular processes

Insertion: Opposite articular process forming joint capsule

Function: Encloses synovial facet joint; limits excessive translation and rotation

Biomechanics: Tension in extension and rotation; capsule fold/meniscoid entrapment theorized in acute facet lock

Load Capacity: Moderate; sprain with whiplash-type or extension-rotation injury

Blood Supply: Medial branch vascular supply

Healing: Synovial inflammation may persist — facet arthropathy

Clinical Tests: Extension-rotation Kemp; PA pillar tenderness; medial branch block diagnostic

Common Injuries: Facet capsule sprain; synovitis; degenerative capsular thickening

MRI: Capsular edema if acute; facet joint effusion

Ultrasound: Not standard

Treatment: Manual therapy mobilization; anti-inflammatory; RF ablation chronic confirmed facet pain

Rehabilitation: Flexion preference if extension intolerant; core and hip mobility

Return to Sport: Extension-rotation without pain

References: Bogduk 2012; Cohen facet intervention reviews

5. Lumbar Muscles

RECORD: Multifidus

ID: MUS-LUM-001

Name: Multifidus

Origin: Posterior sacrum, PSIS, mamillary processes of lumbar vertebrae, transverse processes of thoracic vertebrae

Insertion: Spinous processes 2-4 segments cranially

Innervation: Medial branch dorsal rami of spinal nerves (segmental)

Blood Supply: Segmental arteries and dorsal rami

Fiber Direction: Oblique upward and medial (short segmental stabilizer)

Primary Action: Segmental extension and rotation control of lumbar spine

Secondary Action: Contralateral rotation; stabilizes vertebrae during limb movement

Stabilizing Function: Deep local stabilizer; anticipatory activation before limb movement; atrophy with LBP

Length-Tension: Optimal in neutral lordosis; inhibited with pain and segmental instability

EMG: High activation during rotation, extension, and anticipatory postural tasks; reduced in chronic LBP

Trigger Points: Deep paraspinal tenderness; may refer locally along spine

Pain Referral: Local axial LBP; not typically distal radicular pattern

Stretch: Not primary stretch target; focus on activation

Strengthening: Bird dog, side plank with posterior pelvic tilt, prone leg lifts with neutral spine, segmental co-contraction cues

Motor Control: Re-education early in LBP rehab; co-activation with transversus abdominis

Clinical Importance: Multifidus atrophy correlates with recurrent LBP; target in core stability programs

Common Injuries: Strain with rotation/lifting; inhibition post-disc injury

Rehabilitation: Low-load isometric → dynamic stabilization → functional integration

References: Bogduk 2012; Richardson et al. therapeutic exercise for spinal stabilization; APTA LBP CPG

RECORD: Rotatores

ID: MUS-LUM-002

Name: Rotatores

Origin: Transverse processes

Insertion: Lamina and spinous process of vertebrae 1-2 segments above

Innervation: Medial branch dorsal rami

Blood Supply: Segmental spinal branches

Fiber Direction: Oblique ascending

Primary Action: Segmental rotation and extension

Secondary Action: Fine-tuning intersegmental motion

Stabilizing Function: Deep intrinsic stabilizer with multifidus

Length-Tension: Short muscles — length changes minimal per segment

EMG: Moderate during rotation tasks

Trigger Points: Deep paraspinal

Pain Referral: Local

Stretch: Indirect via global rotation mobility

Strengthening: Rotational bird dog, controlled segmental rotation

Motor Control: Integrated with multifidus training

Clinical Importance: Less commonly isolated in clinical practice vs multifidus

Common Injuries: Rare isolated strain

Rehabilitation: Part of deep stabilizer program

References: Moore; Gray's Anatomy

RECORD: Interspinales

ID: MUS-LUM-003

Name: Interspinales

Origin: Spinous process

Insertion: Spinous process immediately above

Innervation: Medial branch dorsal rami

Blood Supply: Segmental

Fiber Direction: Vertical between adjacent SPs

Primary Action: Segmental extension

Secondary Action: Assist multifidus

Stabilizing Function: Local extensor

Length-Tension: Neutral spine optimal

EMG: Active in extension

Trigger Points: Midline tenderness possible

Pain Referral: Local axial

Stretch: Flexion-based mobility

Strengthening: Prone extension holds, superman progressions

Motor Control: Co-contraction with global extensors

Clinical Importance: Minor contributor vs erector spinae mass

Common Injuries: Uncommon isolated

Rehabilitation: Extension control training

References: Gray's Anatomy

RECORD: Intertransversarii (Lumbar)

ID: MUS-LUM-004

Name: Intertransversarii (Lumbar)

Origin: Transverse process

Insertion: Transverse process of vertebra above

Innervation: Medial branch dorsal rami

Blood Supply: Segmental

Fiber Direction: Vertical between TPs

Primary Action: Lateral flexion (ipsilateral)

Secondary Action: Stabilization

Stabilizing Function: Segmental lateral control

Length-Tension: Short intersegmental

EMG: Lateral flexion tasks

Trigger Points: Lateral paraspinal

Pain Referral: Local lateral LBP

Stretch: Side bending opposite
Strengthening: Side plank, suitcase carry
Motor Control: QL and lateral chain integration
Clinical Importance: Rarely isolated clinically
Common Injuries: Strain with lateral bending
Rehabilitation: Lateral core stability
References: Moore

RECORD: Erector Spinae — Iliocostalis Lumborum

ID: MUS-LUM-005
Name: Erector Spinae — Iliocostalis Lumborum
Origin: Iliac crest, sacrum, lumbar TPs and angles
Insertion: Angles of lower ribs and cervical transverse processes
Innervation: Dorsal rami of spinal nerves
Blood Supply: Lumbar and intercostal arteries
Fiber Direction: Vertical/oblique superior-lateral
Primary Action: Extension of spine; lateral flexion (unilateral)
Secondary Action: Ipsilateral rotation assistance
Stabilizing Function: Global extensor; eccentric control during flexion
Length-Tension: Lengthened in flexion; shortened in extension
EMG: High in extension, lifting, running
Trigger Points: Paraspinal lateral column; may refer to gluteal region
Pain Referral: Lower back, iliac crest, gluteal
Stretch: Child's pose, knee-to-chest (if flexion tolerated)
Strengthening: Back extension, deadlift pattern, rowing
Motor Control: Co-activation with glutes; hip hinge retraining
Clinical Importance: Common muscle strain in lifting athletes
Common Injuries: Erector strain (weightlifting, golf)
Rehabilitation: Graded extension loading; hip hinge retraining
References: Magee 2014; McGill back biomechanics

RECORD: Erector Spinae — Longissimus Thoracis/Lumborum

ID: MUS-LUM-006
Name: Erector Spinae — Longissimus Thoracis/Lumborum
Origin: Lumbar TP, sacrum, iliac crest
Insertion: Lumbar and thoracic TPs, ribs, thoracic SPs
Innervation: Dorsal rami
Blood Supply: Segmental lumbar arteries
Fiber Direction: Vertical long segments
Primary Action: Spinal extension and lateral flexion
Secondary Action: Head/neck extension (thoracic portion)
Stabilizing Function: Primary global extensor mass
Length-Tension: Flexion lengthens
EMG: High during standing, walking, lifting

Trigger Points: Paraspinal; thoracolumbar junction
Pain Referral: Axial and buttock
Stretch: Flexion-based as tolerated
Strengthening: Back extension, bird dog, deadlift variations
Motor Control: Balance with anterior core
Clinical Importance: Major contributor to lifting mechanics
Common Injuries: Strain with flexion-rotation under load
Rehabilitation: Progressive loading; motor pattern retraining
References: Moore; McGill

RECORD: Quadratus Lumborum (QL)

ID: MUS-LUM-007
Name: Quadratus Lumborum (QL)
Origin: Iliac crest, iliolumbar ligament
Insertion: Transverse processes L1-L4, 12th rib
Innervation: Ventrolateral rami T12-L4
Blood Supply: Lumbar, iliolumbar arteries
Fiber Direction: Vertical from iliac crest to TP/12th rib
Primary Action: Ipsilateral lateral flexion; fixes 12th rib during respiration
Secondary Action: Extension assistance
Stabilizing Function: Lateral stabilizer; single-leg stance
Length-Tension: Shortened with scoliosis/listing
EMG: High in side plank, suitcase carry, single-leg tasks
Trigger Points: Deep lateral LBP; SI and trochanter referral
Pain Referral: Lower back, hip, lateral thigh (somatic)
Stretch: Side-lying over ball, overhead reach opposite
Strengthening: Side plank, suitcase carry, single-leg deadlift
Motor Control: Balance with obliques
Clinical Importance: Frequently implicated in nonspecific LBP
Common Injuries: Strain with lifting and rotation
Rehabilitation: Stretch + strengthen + hip/pelvic stability
References: Travell & Simons; Magee 2014

RECORD: Psoas Major

ID: MUS-LUM-008
Name: Psoas Major
Origin: Transverse processes and bodies L1-L5
Insertion: Lesser trochanter of femur (with iliacus)
Innervation: L1-L3 ventral rami (lumbar plexus)
Blood Supply: Lumbar branch iliolumbar artery
Fiber Direction: Oblique inferior-lateral
Primary Action: Hip flexion; lumbar flexion bilateral
Secondary Action: Anterior pelvic tilt; increases lordosis
Stabilizing Function: Deep hip flexor; spine-LE linkage

Length-Tension: Shortened with prolonged sitting
EMG: High in walking, running, leg raises
Trigger Points: Deep anterior hip/LBP
Pain Referral: Anterior groin, lumbar region
Stretch: Half-kneeling hip flexor stretch, Thomas test
Strengthening: Resisted hip flexion, psoas march
Motor Control: Avoid dominant psoas in sit-ups
Clinical Importance: Hip flexor tightness linked to LBP presentations
Common Injuries: Strain in kicking sports
Rehabilitation: Hip flexor mobility + glute activation
References: Moore; Magee 2014

RECORD: Iliacus

ID: MUS-LUM-009
Name: Iliacus
Origin: Iliac fossa
Insertion: Lesser trochanter (iliopsoas tendon)
Innervation: Femoral nerve (L2-L4)
Blood Supply: Iliac branches internal iliac; iliolumbar artery
Fiber Direction: Fan-shaped to tendon
Primary Action: Hip flexion
Secondary Action: Anterior pelvic tilt
Stabilizing Function: Pelvic-femoral linkage
Length-Tension: Shortened with sitting
EMG: Hip flexion activities
Trigger Points: Groin, anterior hip
Pain Referral: Anterior hip
Stretch: Hip flexor stretches
Strengthening: Hip flexion against resistance
Motor Control: Co-activate with glute in gait
Clinical Importance: Iliopsoas vs intra-articular hip differential
Common Injuries: Contusion; tendinopathy
Rehabilitation: Hip flexor eccentric loading
References: Moore; Netter

RECORD: Latissimus Dorsi (Lumbar Attachments)

ID: MUS-LUM-010
Name: Latissimus Dorsi (Lumbar Attachments)
Origin: SP T7-L5 via thoracolumbar fascia; iliac crest; inferior ribs
Insertion: Intertubercular groove humerus
Innervation: Thoracodorsal nerve (C6-C8)
Blood Supply: Thoracodorsal artery
Fiber Direction: Oblique superior-lateral to humerus
Primary Action: Shoulder extension, adduction, IR

Secondary Action: Trunk extension via TL fascia
Stabilizing Function: UE-pelvis kinetic chain
Length-Tension: Affects lumbar fascia tension
EMG: Pull-ups, rows, swimming
Trigger Points: Mid-scapular, lower rib
Pain Referral: Scapular, lower thoracic
Stretch: Overhead reach, doorway lat stretch
Strengthening: Pull-ups, rows, lat pulldown
Motor Control: Coordinate with glute in contralateral patterns
Clinical Importance: TL fascia via lat affects lumbar mechanics
Common Injuries: Strain with overhead activity
Rehabilitation: Scapular control + core + lat strengthening
References: Anatomy Trains superficial back line; Magee 2014

RECORD: Gluteus Maximus (Kinetic Chain Partner)

ID: MUS-LUM-011
Name: Gluteus Maximus (Kinetic Chain Partner)
Origin: Posterior ilium, sacrum, coccyx, sacrotuberous ligament
Insertion: Gluteal tuberosity; iliotibial tract
Innervation: Inferior gluteal nerve (L5-S2)
Blood Supply: Superior and inferior gluteal arteries
Fiber Direction: Oblique inferior-lateral
Primary Action: Hip extension, external rotation
Secondary Action: Posterior pelvic tilt; SI force closure
Stabilizing Function: Primary hip extensor; reduces lumbar erector dominance
Length-Tension: Inhibited with sitting
EMG: Squat, deadlift, sprint
Trigger Points: Buttock to posterior thigh
Pain Referral: Buttock somatic
Stretch: Figure-4, pigeon pose
Strengthening: Bridge, hip thrust, RDL, squat
Motor Control: Glute activation before spinal loading
Clinical Importance: Essential LBP rehab — hip extension offloads spine
Common Injuries: Sprint strain; tendinopathy
Rehabilitation: Glute activation progressions
References: McGill; APTA LBP CPG

RECORD: Transversus Abdominis

ID: MUS-LUM-012
Name: Transversus Abdominis
Origin: Costal cartilages 7-12, TL fascia, iliac crest, inguinal ligament
Insertion: Linea alba, pubic crest
Innervation: T7-L1 intercostal, iliohypogastric/ilioinguinal
Blood Supply: Lower intercostal, deep circumflex iliac

Fiber Direction: Horizontal (transverse)
Primary Action: Abdominal compression; IAP increase
Secondary Action: Minimal trunk motion
Stabilizing Function: Anticipatory trunk stability; corset with multifidus and pelvic floor
Length-Tension: Optimal at neutral spine
EMG: Low sustained; drawing-in increases selectively
Trigger Points: Deep lower abdominal
Pain Referral: Lower abdominal
Stretch: Activation focus over stretch
Strengthening: Drawing-in, dead bug, plank, Pallof press
Motor Control: Motor control cornerstone (Hodges/Richardson)
Clinical Importance: Delayed activation in LBP
Common Injuries: Rare strain
Rehabilitation: Low-level isometric → functional carry/lift
References: Hodges Richardson; APTA LBP CPG; Cochrane exercise chronic LBP

RECORD: Internal Oblique

ID: MUS-LUM-013
Name: Internal Oblique
Origin: TL fascia, iliac crest, inguinal ligament
Insertion: Linea alba, pubic crest, ribs 10-12
Innervation: T7-L1 intercostal, iliohypogastric/ilioinguinal
Blood Supply: Lower intercostal, lumbar arteries
Fiber Direction: Superior-anterior
Primary Action: Ipsilateral lateral flexion; bilateral flexion; same-side rotation
Secondary Action: IAP; TL fascia tension
Stabilizing Function: Anterolateral stability
Length-Tension: Shortened with lateral shift
EMG: Rotation, anti-rotation tasks
Trigger Points: Lower lateral abdomen
Pain Referral: Groin (differentiate hernia)
Stretch: Side-lying stretch
Strengthening: Side plank, Pallof press, wood chop
Motor Control: Coordinate with deep stabilizers
Clinical Importance: Abdominal wall strain rotation sports
Common Injuries: Oblique strain baseball/golf
Rehabilitation: Graded rotation loading
References: Magee 2014

RECORD: External Oblique

ID: MUS-LUM-014
Name: External Oblique
Origin: External surfaces ribs 5-12
Insertion: Linea alba, pubic tubercle, iliac crest

Innervation: T7-L1 intercostal
Blood Supply: Lower intercostal arteries
Fiber Direction: Inferior-medial
Primary Action: Contralateral rotation; ipsilateral lateral flexion; flexion
Secondary Action: IAP; TL fascia
Stabilizing Function: Global anterolateral mover
Length-Tension: As internal oblique
EMG: Rotation, sit-up, throwing
Trigger Points: Lower ribs, lateral abdomen
Pain Referral: Groin, lateral hip
Stretch: Rotation opposite
Strengthening: Controlled rotation, anti-rotation
Motor Control: Balance bilateral obliques
Clinical Importance: Athletic rotation power and strain
Common Injuries: External oblique strain
Rehabilitation: Progressive rotation with stability
References: Magee 2014

RECORD: Rectus Abdominis

ID: MUS-LUM-015
Name: Rectus Abdominis
Origin: Pubic symphysis and crest
Insertion: Xiphoid and costal cartilages 5-7
Innervation: T7-T12 intercostal
Blood Supply: Superior and inferior epigastric arteries
Fiber Direction: Vertical segmented
Primary Action: Trunk flexion; posterior pelvic tilt
Secondary Action: IAP; pelvic stabilization
Stabilizing Function: Global flexor
Length-Tension: Lengthened in lordosis
EMG: Sit-up, curl-up, leg raise
Trigger Points: Periumbilical
Pain Referral: Midline abdominal
Stretch: Cobra if extension tolerated
Strengthening: McGill curl-up, dead bug
Motor Control: Avoid excessive sit-up volume in LBP
Clinical Importance: Diastasis recti modifications
Common Injuries: Strain forceful sit-up
Rehabilitation: Graded flexion with TA co-contraction
References: McGill curl-up; Magee 2014

6. Lumbar Nerves

RECORD: Nerve Root L1

ID: NERVE-L1

Root: L1

Dermatome: Groin/inguinal region (limited distinct patch; overlaps L2)

Myotome: Hip flexion (iliopsoas — L1-L2 contribution); no isolated reliable manual muscle test

Reflex: No primary deep tendon reflex for L1

Motor: Iliopsoas (with L2-L3); quadratus lumborum partial

Sensory: Upper anterior/medial thigh proximal (overlap with L2)

Entrapment: Rare isolated L1 radiculopathy; upper lumbar disc less common

Clinical Presentation: Groin/anterior thigh pain; weak hip flexion if severe

Neurodynamic Tests: Femoral nerve tension (if L2-L3 overlap); SLR typically negative for L1

Differential Diagnosis: Hip joint pathology, hernia, psoas pathology, upper lumbar facet pain

MRI: Upper lumbar disc/facet; nerve root at L1 foramen

EMG: Iliopsoas, quadriceps if L2-L3 involved

Rehabilitation: Hip flexor strengthening; core stability; address upper lumbar hypomobility if present

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Nerve Root L2

ID: NERVE-L2

Root: L2

Dermatome: Anterior thigh to knee (proximal)

Myotome: Hip flexion (iliopsoas); knee extension (quadriceps contribution L2-L4)

Reflex: No isolated L2 reflex

Motor: Iliopsoas, hip adductors (obturator L2-L4 overlap)

Sensory: Anterior mid-thigh

Entrapment: Lateral recess or foraminal disc at L2-L3 affects L2 root

Clinical Presentation: Anterior thigh pain/numbness; hip flexor weakness

Neurodynamic Tests: Femoral nerve bias (prone knee bend); SLR may be negative

Differential Diagnosis: Meralgia paresthetica (LFCN), hip pathology, femoral neuropathy

MRI: L2-L3 disc/foraminal stenosis

EMG: Iliopsoas, quadriceps

Rehabilitation: Neural mobilization femoral bias; progressive hip/knee strengthening

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Nerve Root L3

ID: NERVE-L3

Root: L3

Dermatome: Anterior thigh and knee; medial knee region

Myotome: Knee extension (quadriceps — primarily L3-L4)

Reflex: Patellar reflex contribution (dominant L4)

Motor: Quadriceps, hip flexors, hip adductors

Sensory: Anterior thigh, medial knee

Entrapment: L3-L4 disc paracentral/foraminal

Clinical Presentation: Anterior thigh pain; knee extension weakness if significant

Neurodynamic Tests: Femoral nerve tension; SLR usually negative unless L4+ involvement

Differential Diagnosis: Quadriceps strain, patellofemoral pain, L4 radiculopathy

MRI: L3-L4 level pathology

EMG: Quadriceps

Rehabilitation: Quadriceps strengthening; patellar mobilization if referred knee pain

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Nerve Root L4

ID: NERVE-L4

Root: L4

Dermatome: Medial leg and medial malleolus (classic L4 distribution)

Myotome: Knee extension (quadriceps); ankle dorsiflexion (tibialis anterior contribution L4-L5)

Reflex: Patellar (quadriceps) reflex — primary L4

Motor: Quadriceps, tibialis anterior (with L5), hip adductors

Sensory: Medial lower leg, medial foot

Entrapment: L4-L5 disc most common paracentral herniation level; lateral recess stenosis

Clinical Presentation: Low back pain with medial leg pain; quadriceps weakness; reduced patellar reflex; CAUDA EQUINA: bilateral leg symptoms, saddle anesthesia, urinary retention/incontinence, progressive bilateral weakness — EMERGENCY surgical referral; not isolated single-root pattern

Neurodynamic Tests: SLR positive if L4/L5 root tension; crossed SLR increases specificity

Differential Diagnosis: L5 radiculopathy (lateral leg), spinal stenosis, hip OA referred pain

MRI: Gold standard for L4 root compression at L4-L5

EMG: Quadriceps, tibialis anterior

Rehabilitation: Directional preference; neural glides; progressive LE strengthening; monitor reflex

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Nerve Root L5

ID: NERVE-L5

Root: L5

Dermatome: Lateral leg, dorsum of foot, great toe (web space)

Myotome: Great toe extension (EHL — primarily L5); hip abduction (gluteus medius L5-S1)

Reflex: No primary deep tendon reflex (medial hamstring reflex variable, not routine)

Motor: Extensor hallucis longus, gluteus medius, tibialis anterior (shared L4-L5)

Sensory: Lateral lower leg, dorsum foot, 1st toe

Entrapment: L5-S1 and L4-L5 discs; foraminal/extraforaminal L5 compression common clinically

Clinical Presentation: Sciatica lateral leg/foot; foot drop component if EHL weak; difficulty heel walk; CAUDA EQUINA: bilateral leg symptoms, saddle anesthesia, urinary retention/incontinence, progressive bilateral weakness — EMERGENCY surgical referral; not isolated single-root pattern

Neurodynamic Tests: SLR, slump test, femoral nerve test negative unless dual level

Differential Diagnosis: Peroneal neuropathy at fibular head (foot drop), SI joint pain, piriformis syndrome

MRI: L4-L5 and L5-S1 disc; L5 nerve root in foramen

EMG: EHL, tibialis anterior, gluteus medius

Rehabilitation: Ankle dorsiflexion/EHL strengthening; hip abduction; avoid prolonged flexion if disc

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Nerve Root S1

ID: NERVE-S1

Root: S1

Dermatome: Posterior leg, lateral foot, heel, 4th-5th toes

Myotome: Plantarflexion (gastrocnemius/soleus — S1); hip extension (gluteus maximus)

Reflex: Achilles (ankle jerk) reflex — primary S1

Motor: Gastrocnemius, soleus, gluteus maximus, hamstrings (partial)

Sensory: Posterior calf, lateral foot, sole

Entrapment: L5-S1 disc herniation; lateral recess at L5-S1; piriformis syndrome (sciatic nerve not root but S1 distribution overlap)

Clinical Presentation: Classic sciatica posterior leg; calf weakness; reduced Achilles reflex; CAUDA EQUINA: bilateral leg symptoms, saddle anesthesia, urinary retention/incontinence, progressive bilateral weakness — EMERGENCY surgical referral; not isolated single-root pattern

Neurodynamic Tests: SLR most sensitive for S1 radiculopathy; slump test; bowstring test

Differential Diagnosis: Hamstring strain, Achilles tendinopathy, SI joint, vascular claudication

MRI: L5-S1 disc and S1 root; differentiate from canal stenosis

EMG: Gastrocnemius, gluteus maximus

Rehabilitation: Plantarflexor and hip extension strengthening; walking program; monitor reflex recovery

References: Netter dermatomes/myotomes; Magee 2014; Bogduk 2012 nerve root anatomy

RECORD: Cauda Equina Syndrome (Urgent Pathology Entity)

ID: NERVE-CAUDA-001

Root: Cauda equina (L2-L5 nerve roots within thecal sac below conus medullaris ~L1-L2)

Dermatome: Saddle area (perianal, genitals, medial buttocks) — bilateral

Myotome: Multiple root levels — bilateral leg weakness

Reflex: May lose ankle and patellar reflexes bilaterally

Motor: Bilateral lower extremity weakness; foot drop possible; reduced anal sphincter tone

Sensory: Saddle anesthesia/paresthesia; bilateral leg sensory loss

Entrapment: Large central disc herniation, tumor, epidural abscess, hematoma, fracture fragment

Clinical Presentation: URGENCIA MÉDICA: urinary retention or overflow incontinence, fecal incontinence, saddle anesthesia, bilateral sciatica, progressive neurological deficit; may have back pain

Neurodynamic Tests: Do NOT delay imaging for exhaustive testing — urgent MRI if suspected

Differential Diagnosis: Must distinguish from bilateral radiculopathy, spinal stenosis (usually gradual), conus medullaris syndrome (upper motor neuron signs, earlier bladder involvement at T12-L2)

MRI: EMERGENCY MRI whole lumbar spine with contrast if infection/malignancy suspected

EMG: Not for acute diagnosis — surgical emergency takes priority

Rehabilitation: Post-surgical rehab only after decompression; bowel/bladder rehab; long-term may need multidisciplinary care

References: NICE NG59 red flags; ACR appropriateness MRI lumbar; Bogduk 2012; APTA LBP CPG emergency referral

7. Blood Supply

RECORD: Abdominal Aorta — Lumbar Segment

ID: VAS-LUM-001

Artery: Abdominal Aorta (lumbar segment)

Origin: Continuation of thoracic aorta through diaphragm aortic hiatus (~T12)

Course: Retroperitoneal anterior to lumbar vertebral bodies; bifurcates L4-L5 into common iliac arteries

Branches: Four paired lumbar arteries (L1-L4 typical); middle sacral artery from posterior aspect

Structures Supplied: Posterior abdominal wall, vertebral bodies via segmental branches, spinal cord/cauda equina via radicular arteries

Venous Drainage: Paired lumbar veins to inferior vena cava; ascending lumbar veins anastomose

Clinical Importance: Aortic pathology (aneurysm) may present as back pain; surgical approach awareness

Imaging: CT angiography for vascular pathology; MRI for soft tissue

References: Moore; Gray's Anatomy; Netter

RECORD: Lumbar Arteries (Segmental)

ID: VAS-LUM-002

Artery: Lumbar Arteries (Segmental — typically 4 pairs)

Origin: Posterolateral abdominal aorta at each lumbar level

Course: Pass posteriorly along vertebral body; divide into spinal, psoas, and dorsal branches

Branches: Spinal branch enters intervertebral foramen; psoas branch; dorsal branch to paraspinal muscles

Structures Supplied: Vertebral body, pedicle, paraspinal muscles, psoas, dura, nerve roots

Venous Drainage: Companion lumbar veins; internal vertebral venous plexus communication

Clinical Importance: Bleeding risk in anterior lumbar surgery; segmental artery ligation considerations

Imaging: CT for preoperative vascular mapping

References: Bogduk 2012; surgical anatomy texts

RECORD: Spinal/Radicular Arteries

ID: VAS-LUM-003

Artery: Radicular and Medullary Arteries (Variable)

Origin: Spinal branches of segmental arteries entering IVF

Course: Accompany nerve roots; anterior and posterior radicular arteries; great anterior radiculomedullary artery (artery of Adamkiewicz) typically T9-L2

Branches: Supply nerve roots and contribute to cauda equina/cord perfusion

Structures Supplied: Nerve roots, dura, cauda equina; cord below T8 via Adamkiewicz

Venous Drainage: Radicular veins to internal vertebral plexus

Clinical Importance: Iatrogenic injury during surgery rare but catastrophic (spinal cord/cauda ischemia); collateral circulation variable

Imaging: Angiography rarely preoperative unless vascular malformation suspected

References: Bogduk 2012; Netter neurovascular

RECORD: Iliolumbar Artery

ID: VAS-LUM-004

Artery: Iliolumbar Artery

Origin: Posterior division of internal iliac artery (or common iliac variant)

Course: Ascends behind psoas to iliac fossa; iliolumbar ligament region

Branches: Spinal, iliac, lumbar muscular branches

Structures Supplied: Iliolumbar ligament, QL, psoas, L5 vertebral body region, anastomosis with lumbar arteries

Venous Drainage: Iliolumbar vein to internal iliac system

Clinical Importance: L5 vascular supply; relevant in lumbosacral fusion and posterior approaches

Imaging: CT angiography if vascular anomaly suspected

References: Moore; Bogduk 2012

RECORD: Middle Sacral Artery

ID: VAS-LUM-005

Artery: Middle Sacral Artery

Origin: Posterior aspect of aortic bifurcation

Course: Descends anterior to sacrum to coccyx

Branches: Lateral sacral branches anastomose with iliolumbar and lateral sacral arteries

Structures Supplied: Sacrum, rectum (anterior branches), coccygeal region

Venous Drainage: Middle sacral vein

Clinical Importance: Sacral surgery hemostasis; anastomotic network at lumbosacral junction

Imaging: CT for pelvic vascular anatomy

References: Gray's Anatomy

RECORD: Internal Vertebral Venous Plexus (Batson)

ID: VAS-LUM-006

Artery: Internal Vertebral Venous Plexus (Batson's Plexus) — Venous

Origin: Valveless venous network within vertebral canal and epidural space

Course: Communicates cranially with intracranial sinuses; caudally to sacral plexus; laterally via intervertebral veins to azygos/segmental systems

Branches: Basivertebral veins from vertebral bodies; anterior and posterior plexuses

Structures Supplied: N/A (venous drainage of vertebrae, epidural space, nerve roots)

Venous Drainage: Route for metastatic spread (prostate, breast, lung); engorgement with Valsalva and stenosis

Clinical Importance: Epidural venous engorgement contributes to neurogenic claudication; metastatic workup if cancer history and new back pain

Imaging: MRI shows epidural vein prominence; bone scan/CT for metastases

References: Batson vertebral venous plexus classic description; Moore

RECORD: Ascending Lumbar Veins

ID: VAS-LUM-007

Artery: Ascending Lumbar Veins — Venous

Origin: Common iliac and lumbar venous tributaries

Course: Ascend on anterior vertebral bodies; connect IVC and azygos systems

Branches: Anastomoses with lumbar veins and vertebral plexus

Structures Supplied: Venous drainage posterior abdominal wall and vertebrae

Venous Drainage: Collateral pathway if IVC obstruction

Clinical Importance: Surgical landmark in retroperitoneal approaches

Imaging: CT venography if indicated

References: Gray's Anatomy

8. Lumbar Fascia

RECORD: Thoracolumbar Fascia

ID: FAS-LUM-001

Structure: Thoracolumbar Fascia (TLF)

Attachments: Spinous processes, iliac crest, ribs, latissimus dorsi, internal oblique, transversus abdominis, sacroiliac region; three layers (posterior, middle, anterior) investing QL

Continuity: Continuous with thoracic fascia superiorly, gluteal fascia inferiorly, abdominal fascia anteriorly; integrates with posterior layer of abdominal wall

Force Transmission: Transfers loads between spine, pelvis, and lower extremity; eccentric load during lifting via latissimus and glute connections; fascial tensioning with IAP

Clinical Importance: Central to lumbopelvic stability models; TLF dysfunction implicated in LBP and lifting mechanics

Restrictions: Thickened/fibrotic fascia may limit trunk mobility; post-surgical scarring

Assessment: Palpation paraspinal and lateral to QL; fascial glide assessment in side-lying; observe lifting pattern and fascial tension with bracing

Treatment: Myofascial release, instrument-assisted soft tissue mobilization (limited evidence), loading through functional patterns preferred over passive-only

Evidence: Fascial role in force transfer supported biomechanically; manual fascial release evidence modest — exercise and loading stronger evidence base (APTA LBP CPG)

References: Myers Anatomy Trains; Bogduk 2012; Willard TLF anatomy research

RECORD: Lumbar Fascia (Paraspinal/Investing)

ID: FAS-LUM-002

Structure: Lumbar Fascia (Paraspinal and Investing Layers)

Attachments: Invests erector spinae; attaches to lumbar SP, TP, and iliac crest

Continuity: Part of TLF system; separates paraspinal compartment

Force Transmission: Compartmental pressure during extension and lifting; distributes erector spinae force

Clinical Importance: Paraspinal compartment concept in back muscle strain; palpation tenderness in myofascial LBP

Restrictions: Muscle guarding increases fascial tension; chronic pain central sensitization overlap

Assessment: Palpation paraspinal muscle and overlying fascia; tone at rest vs activation

Treatment: Heat, gentle mobilization, progressive loading, dry needling (where permitted) for trigger points

Evidence: Active exercise superior to passive-only for chronic LBP (Cochrane reviews)

References: Gray's Anatomy; Travell & Simons

RECORD: Abdominal Fascia (Anterior/Lateral Abdominal Wall)

ID: FAS-LUM-003

Structure: Abdominal Fascia (Anterior and Lateral Wall)

Attachments: Rectus sheath, linea alba, external/internal oblique aponeuroses, inguinal ligament, iliac crest

Continuity: Integrates with TLF posteriorly; forms rectus sheath enwrapping rectus abdominis

Force Transmission: Transfers IAP and anterior tension; oblique lines of force to pelvis and spine

Clinical Importance: Core stability — fascial tension with TA/oblique co-contraction; hernia differential in groin pain

Restrictions: Post-surgical (C-section, appendectomy) adhesions may alter motor control

Assessment: Abdominal bracing observation; diastasis recti gap measurement; cough impulse

Treatment: Progressive core training; address diastasis with appropriate progressions

Evidence: Motor control and graded exercise for LBP — moderate-strong recommendation (NICE NG59, APTA CPG)

References: Moore; Hodges/Richardson core stability research

9. Biomechanics

RECORD: Flexion

ID: BIO-LUM-001

Movement: Lumbar Flexion

Plane: Sagittal

Axis: Coronal axis (mediolateral)

Primary Movers: Rectus abdominis, psoas (bilateral), external obliques

Stabilizers: Erector spinae eccentric control; multifidus; transversus abdominis

Arthrokinematics: Superior facets slide anterior-superior; disc nucleus migrates posteriorly; posterior annulus tension increases

Joint Reaction Forces: Posterior annulus and PLL loaded; intradiscal pressure increases with flexion load

Torque: Flexion moment increases with forward bending and anterior load (lifting far from body)

Length-Tension: Erector spinae lengthened; may reduce passive tension capacity if held prolonged flexion

EMG: Abdominals increase; erector spinae eccentric activity in controlled flexion

Compensations: Hip hinge loss → excessive lumbar flexion in lifting; thoracic kyphosis compensation

Common Dysfunctions: Flexion-intolerant LBP (disc, facet in some); flexion-relief stenosis pattern opposite

Clinical Relevance: McKenzie extension vs flexion preference; sit-to-stand pattern assessment

Assessment: Repeated flexion/extension testing; finger-to-floor; sit-slouch test

Exercises: Cat-camel, curl-up (McGill), hip hinge drill, dead bug

References: McGill; Nachemson intradiscal pressure; Bogduk 2012

RECORD: Extension

ID: BIO-LUM-002

Movement: Lumbar Extension

Plane: Sagittal

Axis: Coronal axis

Primary Movers: Erector spinae, multifidus, QL (assist)

Stabilizers: Abdominals eccentric; gluteus maximus posterior pelvic tilt assist

Arthrokinematics: Inferior facets slide posterior-inferior; foraminal narrowing; ligamentum flavum buckling

Joint Reaction Forces: Facet joints load-share increases (~30% approx in extension); pars stress in hyperextension athletes

Torque: Extension moment in standing backbend, overhead reach, running uphill

Length-Tension: Erector spinae shortened; abdominals lengthened

EMG: High erector activity in prone extension, standing extension

Compensations: Hyperlordosis — dominant lumbar extension vs hip extension; spondylolytic athlete extension pain

Common Dysfunctions: Extension-intolerant LBP (facet syndrome, stenosis); spondylolysis pain with extension/rotation

Clinical Relevance: Stenosis worse extension-standing; facet syndrome extension provocative

Assessment: Prone press-up; standing extension; repeated extension test

Exercises: Prone props (McKenzie), hip extension (bridge), avoid end-range extension in stenosis

References: Bogduk 2012; Magee 2014

RECORD: Rotation

ID: BIO-LUM-003

Movement: Lumbar Rotation

Plane: Transverse

Axis: Vertical (superior-inferior)

Primary Movers: Internal/external obliques, multifidus, rotatores

Stabilizers: Transversus abdominis; contralateral QL

Arthrokinematics: Minimal lumbar rotation (~1-2 deg/segment); coupled with lateral flexion

Joint Reaction Forces: Torsional stress on disc annulus; pars stress in rotation-extension (cricket, golf)

Torque: Rotation torque in throwing, golf swing, shoveling

Length-Tension: Obliques alternately shortened/lengthened

EMG: High oblique and paraspinal during rotation sports

Compensations: Thoracic rotation compensation if lumbar stiff; excessive lumbar rotation in golf

Common Dysfunctions: Disc annular tear with rotation-flexion; spondylolysis in rotation sports

Clinical Relevance: Modify rotation load in acute disc injury; golf swing biomechanics

Assessment: Seated rotation ROM; rotation with extension Kemp test

Exercises: Thoracic rotation mobility, anti-rotation Pallof press, controlled wood chop progressions

References: McGill; sports biomechanics literature

RECORD: Side Bending (Lateral Flexion)

ID: BIO-LUM-004

Movement: Side Bending (Lateral Flexion)

Plane: Frontal

Axis: Anteroposterior axis

Primary Movers: Quadratus lumborum, erector spinae (unilateral), obliques

Stabilizers: Contralateral QL and obliques; multifidus

Arthrokinematics: Ipsilateral facets close; contralateral gapping; intertransverse ligament tension

Joint Reaction Forces: Asymmetric disc loading; iliolumbar ligament stress

Torque: Lateral bend with carrying load on one side (suitcase)

Length-Tension: Ipsilateral QL shortened; contralateral lengthened

EMG: QL and obliques high in side plank, lateral carry

Compensations: Listing antalgic posture in radiculopathy; scoliosis structural vs functional

Common Dysfunctions: QL strain; disc herniation with lateral shift (McKenzie correction)

Clinical Relevance: Antalgic list reduction via extension or side glide techniques

Assessment: Standing lateral flexion; observe list; side glide response

Exercises: Side plank, suitcase carry, controlled lateral flexion mobility

References: Magee 2014; McKenzie mechanical diagnosis

RECORD: Lifting

ID: BIO-LUM-005

Movement: Lifting (Manual Material Handling)

Plane: Multi-planar (primarily sagittal)

Axis: Multiple

Primary Movers: Gluteus maximus, hamstrings, quadriceps, erector spinae

Stabilizers: Transversus abdominis, multifidus, IAP, TL fascia

Arthrokinematics: Hip hinge preserves neutral lumbar lordosis; flexion increases disc pressure

Joint Reaction Forces: Intradiscal pressure highest with flexion + compression + anterior load (Nachemson classic studies)

Torque: Moment arm of load critical — keep load close to body

Length-Tension: Glute/hamstring length in hip hinge; erector co-contraction

EMG: Glute and erector high; abdominals bracing pre-lift

Compensations: Leg-dominant vs back-dominant lifter; stoop lift vs squat lift patterns

Common Dysfunctions: Disc injury with flexion-rotation under load; erector strain

Clinical Relevance: Workplace ergonomics; return-to-work lifting assessment

Assessment: Observe lifting technique; load tolerance testing graduated

Exercises: Hip hinge, deadlift progression, farmer carry, bracing drills

References: McGill Big 3; NIOSH lifting equation; APTA RTW guidelines

RECORD: Sitting

ID: BIO-LUM-006

Movement: Sitting

Plane: Sagittal (postural)

Axis: Coronal

Primary Movers: Minimal active — postural muscles isometric

Stabilizers: Erector spinae, multifidus, abdominals maintain lordosis

Arthrokinematics: Flexed sitting flattens lordosis; increased posterior disc pressure

Joint Reaction Forces: Intradiscal pressure elevated vs standing (slouched sitting worst per classic studies)

Torque: Sustained flexion creep in disc viscoelastic tissues

Length-Tension: Hip flexors shortened; glutes inhibited

EMG: Low-level paraspinal; increased with unsupported sitting

Compensations: Posterior pelvic tilt slouch; perching on seat edge

Common Dysfunctions: Prolonged sitting LBP risk factor (multifactorial)

Clinical Relevance: Micro-breaks, lumbar support, sit-stand desks

Assessment: Workstation ergonomics; sitting tolerance

Exercises: Hip flexor stretch, glute activation, postural breaks

References: NICE NG59 stay active; occupational health guidelines

RECORD: Walking

ID: BIO-LUM-007

Movement: Walking (Gait)

Plane: Sagittal primary with transverse rotation

Axis: Multiple

Primary Movers: Hip flexors/extensors, erector spinae, abdominals rhythmically

Stabilizers: Multifidus, QL, gluteus medius

Arthrokinematics: Pelvic rotation and lumbar counter-rotation (~2-4 deg)

Joint Reaction Forces: Moderate cyclic loading; beneficial disc nutrition pumping

Torque: Alternating hip moments

Length-Tension: Rhythmic muscle length changes

EMG: Erector and abdominals alternate; glute max at terminal stance

Compensations: Reduced arm swing; shortened stride in LBP; flexed posture in stenosis
Common Dysfunctions: Neurogenic claudication limits walking distance in stenosis
Clinical Relevance: First-line aerobic activity for LBP (NICE); graded walking programs
Assessment: Gait observation; walking tolerance; distance to symptom onset
Exercises: Graded walking progression; treadmill incline for stenosis (flexed posture)
References: NICE NG59; Cochrane walking exercise LBP

RECORD: Running

ID: BIO-LUM-008

Movement: Running

Plane: Sagittal with transverse components

Axis: Multiple

Primary Movers: Hip extensors, quadriceps, erector spinae, abdominals

Stabilizers: Multifidus, gluteus medius, core

Arthrokinematics: Higher impact than walking; lumbar lordosis maintained by co-contraction

Joint Reaction Forces: Ground reaction force 2-3× body weight approx; disc cyclical loading

Torque: Increased trunk stabilization demand at faster speeds

Length-Tension: Hip flexor tightness affects stride and lumbar posture

EMG: Higher paraspinal and glute than walking

Compensations: Overstriding; excessive lumbar extension at toe-off; pelvic drop

Common Dysfunctions: Disc and facet irritation with high impact if not conditioned

Clinical Relevance: Return-to-run criteria after LBP; gradual load progression

Assessment: Running gait analysis; symptom response 24h post-run

Exercises: Bridge, single-leg stability, gradual run-walk progression, cadence modification

References: Sports medicine return-to-run protocols; McGill spine stability for athletes

10. Pathologies

RECORD: Disc Herniation

ID: PATH-LUM-001

Condition: Lumbar Disc Herniation (Protrusion/Extrusion/Sequestration)

Definition: Displacement of disc material beyond intervertebral disc space — bulge, protrusion, extrusion, or sequestration per standardized nomenclature

Mechanism: Annular tear with nucleus migration; often flexion-rotation loading; degeneration predisposes

Symptoms: Axial LBP ± radicular leg pain below knee; dermatomal paresthesia; worse with flexion/sit/Valsalva

Red Flags: Cauda equina (bilateral symptoms, saddle anesthesia, urinary retention), progressive motor deficit, fever, cancer history

Differential Diagnosis: Radiculopathy other causes, facet syndrome, SI joint, hip pathology, hamstring strain

Clinical Tests: SLR (sensitive sciatica), crossed SLR (specific), slump test, neurological exam myotome/reflex

Imaging: MRI gold standard; X-ray excludes other pathology; CT for bony detail

Healing: Many improve conservatively 6-12 weeks; sequestration may resolve or need surgery

Treatment: Stay active, education, NSAIDs, PT; epidural injection selective; microdiscectomy if persistent radiculopathy or cauda equina

Manual Therapy: McKenzie directional preference; neural mobilization; mobilization per tolerance

Exercises: Directional preference exercises; core stability; walking; avoid provocative flexion if intolerant

Return to Activity: Gradual loading when leg pain resolving, strength normalized, no neuro progression

Evidence: NICE NG59 conservative first; surgery faster leg pain relief short-term, similar long-term many cases (SPORT trial concept)

References: NICE NG59; APTA LBP CPG; Jensen NEJM disc imaging; Bogduk 2012

RECORD: Lumbar Stenosis

ID: PATH-LUM-002

Condition: Lumbar Spinal Stenosis

Definition: Narrowing of spinal canal, lateral recess, or foramen causing neural compression — central, lateral recess, or foraminal

Mechanism: Degenerative — disc height loss, facet hypertrophy, ligamentum flavum buckling; congenital narrow canal

Symptoms: Neurogenic claudication — bilateral leg pain/heaviness with walking/standing, relieved by sitting/flexion (shopping cart sign)

Red Flags: Acute cauda equina, progressive bilateral weakness, bowel/bladder dysfunction

Differential Diagnosis: Vascular claudication (pulses, ABI), hip OA, peripheral neuropathy, radiculopathy

Clinical Tests: Walking tolerance; symptom relief with lumbar flexion; bicycle test (flexed) vs treadmill (extended)

Imaging: MRI or CT myelography; Schizas grade for canal stenosis; standing imaging if dynamic stenosis suspected

Healing: Chronic progressive degenerative condition; symptoms fluctuate

Treatment: PT flexion-biased exercises, activity modification; NSAIDs; epidural injections; decompression surgery if refractory

Manual Therapy: Flexion mobilization; avoid extension HVLA; gentle neural glides

Exercises: Cycling, flexion-based core, hip flexor strengthening, stationary bike, partial curl-up

Return to Activity: Walking distance goals with flexion relief strategies; pacing

Evidence: Moderate evidence for PT and surgical decompression for moderate-severe symptomatic stenosis (Cochrane/systematic reviews)

References: NICE NG59; North American Spine Society stenosis guidelines; Magee 2014

RECORD: Spondylolysis

ID: PATH-LUM-003

Condition: Spondylolysis (Pars Interarticularis Defect)

Definition: Stress fracture or defect in pars interarticularis without vertebral slippage

Mechanism: Repetitive extension-rotation loading in adolescents (gymnastics, cricket, football linemen); L5 most common

Symptoms: Extension-intolerant LBP; localized paraspinal pain; may mimic disc pain in young athlete

Red Flags: Night pain, fever, weight loss (tumor/infection), neurological deficit (unusual isolated pars)

Differential Diagnosis: Disc injury, facet sprain, SI joint, muscle strain in adolescent athlete

Clinical Tests: One-legged hyperextension test (Stork test — low sensitivity); extension-rotation pain; palpation L5-S1

Imaging: Oblique X-ray (Scottie dog collar defect); MRI (bone edema early); SPECT/CT for active vs chronic

Healing: Acute: 3-6 months bracing and activity modification may heal; chronic fibrous non-union may be asymptomatic

Treatment: Relative rest from extension sports; bracing acute phase; PT core and hip; bone stimulator selective cases

Manual Therapy: Avoid extension manipulation; flexion-based mobilization; soft tissue

Exercises: McGill Big 3, glute/hamstring strengthening, anti-extension core, gradual return

Return to Activity: Pain-free sport-specific drills; no extension pain; imaging stability optional

Evidence: Activity modification and rehabilitation mainstay; surgical fusion rare unilateral active defect failing conservative care

References: Magee 2014 adolescent spine; Bogduk 2012; sports medicine pars literature

RECORD: Spondylolisthesis

ID: PATH-LUM-004

Condition: Spondylolisthesis (Vertebral Slippage)

Definition: Anterior slippage of vertebral body on subjacent — isthmic (pars defect), degenerative, traumatic, dysplastic

Mechanism: Isthmic: pars bilateral defect (often L5 on S1); degenerative: facet arthritis L4-L5 common

Symptoms: LBP, hamstring tightness, radiculopathy if nerve compressed; step-off palpable severe cases

Red Flags: Progressive slip, cauda equina, bowel/bladder dysfunction, high-grade slip in adolescent

Differential Diagnosis: Disc herniation, stenosis, SI joint; differentiate isthmic vs degenerative by age/imaging

Clinical Tests: Meyerding grade on lateral X-ray; Phalen-Dickson sign (flexed knee gait); neuro exam

Imaging: Standing lateral X-ray (flexion-extension views if instability suspected); MRI for nerve compression

Healing: Low-grade isthmic may stabilize; high-grade may progress in adolescents

Treatment: Conservative PT core stabilization for low-grade; fusion for high-grade, progressive slip, or refractory neuro symptoms

Manual Therapy: Stabilization focus; avoid aggressive extension manipulation high-grade

Exercises: Multifidus/TA training, hip strengthening, neutral spine functional training

Return to Activity: Sport-dependent; low-grade may return with core conditioning; monitor symptoms

Evidence: Conservative care first low-grade; surgical fusion for progression or neuro compression (guideline consensus)

References: Bogduk 2012; Wiltse classification; Magee 2014

RECORD: Facet Syndrome

ID: PATH-LUM-005

Condition: Lumbar Facet Syndrome (Zygapophyseal Joint Pain)

Definition: Pain arising from lumbar facet joints and capsules — clinical diagnosis supported by response to diagnostic blocks

Mechanism: Degeneration, synovitis, capsular strain, extension-rotation injury; age-related arthropathy

Symptoms: Deep axial LBP; buttock referral; extension-rotation provocative; morning stiffness

Red Flags: Standard red flags apply; fracture, infection, malignancy if trauma/systemic symptoms

Differential Diagnosis: Discogenic pain, SI joint, myofascial pain, radiculopathy (below knee suggests nerve root)

Clinical Tests: Extension-rotation Kemp; PA pillar tenderness; diagnostic medial branch block (research threshold $\geq 80\%$ relief)

Imaging: X-ray/MRI facet arthropathy common age-related; imaging correlates poorly with pain

Healing: Fluctuating chronic course; acute capsular sprain 4-8 weeks

Treatment: PT flexion-biased, NSAIDs, medial branch block, radiofrequency ablation chronic confirmed cases

Manual Therapy: Mobilization grade I-IV; flexion distraction; avoid HVLA extension stenosis

Exercises: Flexion preference, core stability, hip mobility, postural education

Return to Activity: Extension-rotation without pain; functional lifting with neutral spine

Evidence: RF ablation moderate evidence short-medium term for confirmed facet pain (Cohen systematic reviews)

References: Bogduk 2012 facet innervation; Cohen facet interventions; APTA LBP CPG

RECORD: SI Referral (Sacroiliac Joint Related Pain)

ID: PATH-LUM-006

Condition: Sacroiliac Joint Related Low Back Pain (Referral Pattern)

Definition: Pain arising from or referring through sacroiliac joint region — often presents as lower lumbar/buttock pain

Mechanism: Joint dysfunction, ligamentous strain (iliolumbar), pregnancy hormones, trauma, leg length asymmetry

Symptoms: Unilateral buttock pain below L5, groin referral possible; sit-to-stand pain; PSIS tenderness

Red Flags: Inflammatory arthritis (AS), infection, fracture, malignancy, cauda equina (different pattern)

Differential Diagnosis: L5-S1 radiculopathy, facet pain, hip OA, piriformis syndrome, disc herniation

Clinical Tests: Cluster: distraction, thigh thrust, FABER, compression, Gaenslen (3+ positive suggests SI — modest specificity)

Imaging: X-ray/MRI often normal; bone scan/CT for inflammatory or fracture workup

Healing: Acute sprain 4-8 weeks; chronic may need stabilization program

Treatment: Manual therapy SI mobilization; exercise lumbopelvic stability; injection if persistent

Manual Therapy: Muscle energy, SI mobilization, soft tissue QL/piriformis

Exercises: Bridge, clamshell, single-leg stability, hip hinge retraining

Return to Activity: Single-leg tasks pain-free; sport-specific loading

Evidence: Multimodal PT and manual therapy modest benefit; diagnostic injection gold standard for research

References: Magee 2014 SI tests; APTA; Laslett SI cluster research

RECORD: Muscle Strain (Lumbar Paraspinal / QL)

ID: PATH-LUM-007

Condition: Lumbar Muscle Strain (Paraspinal / Quadratus Lumborum)

Definition: Overstretch or tear of lumbar musculature — erector spinae, multifidus, QL common

Mechanism: Sudden lift, rotation, or eccentric overload; deconditioning; poor hip hinge mechanics

Symptoms: Localized LBP worsened by movement; muscle spasm; no radicular pattern below knee typically

Red Flags: If trauma high-energy — fracture; if fever — infection; neuro deficit — radiculopathy not strain

Differential Diagnosis: Disc, facet, SI joint, kidney stones, aortic aneurysm (elderly/vascular risk)

Clinical Tests: Palpation spasm/trigger points; pain with resisted extension/rotation; normal neuro exam

Imaging: Not required initial uncomplicated strain; MRI if not improving 4-6 weeks

Healing: Grade I 1-2 weeks, Grade II 3-6 weeks, Grade III months

Treatment: Relative rest not bed rest, heat, NSAIDs, progressive mobility and strengthening

Manual Therapy: Soft tissue, gentle mobilization, dry needling where available

Exercises: Graded walking, McGill Big 3, hip hinge retraining, return to lifting progression

Return to Activity: Full ROM and strength without spasm; sport when functional tests pass

Evidence: Active recovery superior to bed rest (Cochrane); early PT beneficial

References: NICE NG59; APTA LBP CPG; Magee 2014

RECORD: Radiculopathy

ID: PATH-LUM-008

Condition: Lumbar Radiculopathy

Definition: Nerve root pathology causing dermatomal pain, paresthesia, weakness, reflex changes

Mechanism: Disc herniation, foraminal stenosis, lateral recess stenosis, synovial cyst compressing root

Symptoms: Leg pain below knee often > back pain; dermatomal sensory loss; myotome weakness; reflex diminished

Red Flags: Cauda equina, progressive deficit, bilateral symptoms, sphincter dysfunction — EMERGENCY

Differential Diagnosis: Peripheral nerve entrapment (sciatic, peroneal), hip pathology, vascular claudication

Clinical Tests: SLR, crossed SLR, slump, femoral stretch L2-L4; myotome/reflex/sensation mapping

Imaging: MRI nerve root compression correlation; EMG/NCV subacute/chronic or if surgery planned

Healing: 80% improve non-surgically within 3 months many cases; depends on compression severity

Treatment: Conservative PT, medications, epidural steroid; surgery if progressive or cauda equina

Manual Therapy: Neural mobilization, directional preference, centralization techniques

Exercises: Core stability, nerve glides, progressive LE strengthening, walking

Return to Activity: When neuro exam stable/improving, leg pain centralizing or resolving

Evidence: NICE NG59; SPORT trial concepts — surgery vs conservative similar long-term many patients

References: NICE NG59; APTA LBP CPG; Magee 2014 neuro exam

RECORD: Cauda Equina Syndrome

ID: PATH-LUM-009

Condition: Cauda Equina Syndrome (CES) — EMERGENCY

Definition: Compression of cauda equina nerve roots (below conus ~L1-L2) causing bilateral neuro deficits and sphincter dysfunction

Mechanism: Large central disc herniation most common; also tumor, hematoma, abscess, fracture, stenosis

Symptoms: Bilateral leg pain/weakness/numbness; saddle anesthesia; urinary retention or incontinence; fecal incontinence; reduced anal tone

Red Flags: ALL ARE RED FLAGS — URGENT SURGICAL REFERRAL WITHIN HOURS: urinary retention, post-void residual >100-200 mL, saddle anesthesia, bilateral sciatica, progressive weakness, fecal incontinence; do not delay MRI for prolonged PT trial

Differential Diagnosis: Conus medullaris syndrome (UMN signs, symmetric, earlier bladder, T12-L2 level); bilateral radiculopathy; peripheral neuropathy (diabetes — gradual not acute)

Clinical Tests: Perianal sensation; anal sphincter tone (PR exam by physician); post-void bladder scan; bilateral lower extremity motor/sensory exam; DO NOT rely on SLR alone

Imaging: EMERGENCY MRI whole lumbar spine — gold standard; if unavailable urgent transfer to center with MRI

Healing: Time-dependent — best outcome surgical decompression within 24-48 hours; delayed may have permanent bladder/bowel/sexual dysfunction

Treatment: EMERGENCY neurosurgical/orthopedic spinal decompression; IV steroids controversial; NOT conservative PT as primary

Manual Therapy: CONTRAINDICATED as primary treatment — urgent referral; PT only post-operative rehabilitation

Exercises: Post-surgical rehab only after decompression per surgical protocol

Return to Activity: Long-term multidisciplinary; bladder training; mobility and strength per residual deficit

Evidence: Strong consensus: CES is surgical emergency; delayed decompression worsens outcomes (systematic reviews/guidelines)

References: NICE NG59 red flags; ACR appropriateness MRI lumbar; Bogduk 2012; APTA emergency referral protocols

11. Rehabilitation Phases

RECORD: Disc Herniation — Acute

ID: REHAB-LUM-001

Condition: Disc Herniation

Phase: Acute

Goals: Reduce leg pain centralization if possible; maintain ADLs; avoid bed rest; screen red flags

Pain Management: NSAIDs per MD; relative activity modification; avoid prolonged flexion if intolerant

Education: Prognosis favorable most cases; stay active; avoid catastrophizing; red flag awareness

Mobility: Directional preference (often extension); gentle walking; avoid SLR aggravation early

Motor Control: Light TA bracing; neutral spine supine; diaphragmatic breathing

Strengthening: Deferred or isometric glute/quad if leg pain allows

Core Stability: McGill curl-up minimal range; side plank modified knees

Functional Training: Sit-stand cues; avoid heavy lifting

Return to Work: Modified duties; no heavy lifting 2-4 weeks typically

Return to Sport: Not acute phase

Progression: Pain centralizing or reducing → subacute loading

Evidence: NICE NG59 stay active; avoid bed rest

References: NICE NG59; McKenzie; APTA LBP CPG

RECORD: Disc Herniation — Subacute

ID: REHAB-LUM-002

Condition: Disc Herniation

Phase: Subacute

Goals: Restore ROM; begin strengthening; normalize gait

Pain Management: Gradual exposure to provocative movements per assessment

Education: Disc healing biology; flare management; posture variability not rigid

Mobility: Repeated extension or flexion per preference; neural glides if radicular

Motor Control: Bird dog, dead bug, TA-multifidus co-contraction

Strengthening: Bridge, clamshell, bodyweight squat pattern

Core Stability: McGill Big 3 progressions

Functional Training: Light lifting technique hip hinge introduction

Return to Work: Gradual increase load tolerance

Return to Sport: Light aerobic (bike, walk)

Progression: Strength 50% baseline → remodeling phase

Evidence: Exercise therapy moderate effect chronic/subacute LBP (Cochrane)

References: Cochrane exercise LBP; McGill

RECORD: Disc Herniation — Remodeling/Functional

ID: REHAB-LUM-003

Condition: Disc Herniation

Phase: Remodeling/Functional

Goals: Full strength; sport/work specific capacity; prevent recurrence

Pain Management: Intermittent soreness acceptable; distinguish from radicular worsening

Education: Self-management; independent program; yellow flags psychosocial screen
Mobility: Full ROM; rotation and extension per tolerance
Motor Control: Dynamic stabilization under load
Strengthening: Deadlift/RDL progression, lunges, row, hip thrust
Core Stability: Loaded carries, Pallof press, anti-rotation
Functional Training: Sport-specific patterns; lifting capacity testing
Return to Work: Full duties if lifting tests passed
Return to Sport: Running/jumping progressions if applicable
Progression: Functional testing → RTS/RTW clearance
Evidence: Graded activity and exercise reduce recurrence risk
References: APTA LBP CPG; sports return-to-play consensus

RECORD: Disc Herniation — RTS/RTW

ID: REHAB-LUM-004

Condition: Disc Herniation

Phase: RTS/RTW

Goals: Full return without symptom limitation; confidence and self-efficacy

Pain Management: Independent flare plan

Education: Maintenance program 2-3×/week

Mobility: Maintain hip and thoracic mobility

Motor Control: Automatic stabilization with sport/work tasks

Strengthening: Continued periodized program

Core Stability: Integrated with athletic training

Functional Training: Full sport drills or job simulation

Return to Work: Full duty achieved

Return to Sport: Full competition; no antalgic movement; neuro exam normal

Progression: Maintenance and periodized deload

Evidence: Active patients have better long-term outcomes

References: NICE NG59; return-to-work guidelines

RECORD: Lumbar Stenosis — Acute

ID: REHAB-LUM-005

Condition: Lumbar Stenosis

Phase: Acute

Goals: Reduce neurogenic claudication symptoms; maintain safe mobility

Pain Management: Flexion relief positions; pacing walking

Education: Stenosis anatomy; flexion opens canal; realistic pacing expectations

Mobility: Knee-to-chest; pelvic tilt; avoid prolonged standing extension

Motor Control: Flexion-biased posture awareness

Strengthening: Seated or flexed-position hip strengthening

Core Stability: Partial curl-up; flexion-based

Functional Training: Shopping cart posture for walking tolerance

Return to Work: Modified standing duties

Return to Sport: Not applicable acute

Progression: Walking distance improving

Evidence: Flexion-based PT supported for stenosis symptoms

References: NASS stenosis guidelines; APTA

RECORD: Lumbar Stenosis — Subacute

ID: REHAB-LUM-006

Condition: Lumbar Stenosis

Phase: Subacute

Goals: Increase walking tolerance; hip flexor strength for flexed posture

Pain Management: Activity pacing; seated rest breaks

Education: Cycle vs walk for exercise; avoid extension exercises initially

Mobility: Hip flexion stretches; thoracic mobility

Motor Control: Abdominal bracing in flexion

Strengthening: Stationary bike; hip flexor/endurance

Core Stability: Flexion-biased core exercises

Functional Training: Graded walking program with flexion breaks

Return to Work: Standing tolerance building

Return to Sport: Cycling, swimming preferred

Progression: Walking >15-20 min with manageable symptoms

Evidence: Moderate evidence supervised exercise stenosis

References: Cochrane stenosis exercise reviews

RECORD: Lumbar Stenosis — Remodeling/Functional

ID: REHAB-LUM-007

Condition: Lumbar Stenosis

Phase: Remodeling/Functional

Goals: Maximize functional walking capacity; hip and core endurance

Pain Management: Intermittent claudication management strategies

Education: Long-term self-management; surgical consultation if refractory

Mobility: Maintain flexion ROM hip and lumbar

Motor Control: Dynamic flexion bias during prolonged standing tasks

Strengthening: Leg press, hip flexion endurance, glute strengthening

Core Stability: Endurance focus over max strength

Functional Training: Simulated work standing tasks with pacing

Return to Work: Full duty with accommodation if needed

Return to Sport: Golf modification (flexed setup); cycling

Progression: Functional goals met or surgical pathway

Evidence: Surgery vs PT — shared decision moderate stenosis

References: NASS; NICE

RECORD: Lumbar Stenosis — RTS/RTW

ID: REHAB-LUM-008

Condition: Lumbar Stenosis

Phase: RTS/RTW

Goals: Sustainable activity level; independence
Pain Management: Self-pacing strategies
Education: Maintenance flexion-biased program
Mobility: Daily hip flexor mobility
Motor Control: Automatic pacing
Strengthening: Maintenance 2-3× weekly
Core Stability: Endurance maintenance
Functional Training: Community walking programs
Return to Work: Achieved with or without modifications
Return to Sport: Low-impact sport maintenance
Progression: Annual review if symptoms progressive
Evidence: Long-term management chronic stenosis
References: North American Spine Society

RECORD: Spondylolysis / Spondylolisthesis — Acute

ID: REHAB-LUM-009

Condition: Spondylolysis / Spondylolisthesis

Phase: Acute

Goals: Pain control; protect pars in acute phase; maintain fitness

Pain Management: Avoid extension/rotation sports; bracing per MD protocol adolescent acute

Education: Pars anatomy; healing timeline 3-6 months; sport modification

Mobility: Flexion-biased gentle ROM; hip mobility

Motor Control: Neutral spine awareness; anti-extension bracing

Strengthening: Isometric glute; avoid loaded extension

Core Stability: McGill Big 3 — curl-up, side plank, bird dog

Functional Training: Non-impact cardio (bike, pool)

Return to Work: Avoid extension-loaded work tasks

Return to Sport: Stop extension sports acutely

Progression: Pain-free daily activities → subacute loading

Evidence: Activity modification key for pars healing adolescents

References: Magee adolescent spine; sports medicine pars protocols

RECORD: Spondylolysis / Spondylolisthesis — Subacute

ID: REHAB-LUM-010

Condition: Spondylolysis / Spondylolisthesis

Phase: Subacute

Goals: Core and hip strengthening; gradual activity reintroduction

Pain Management: Monitor extension pain response

Education: Neutral spine lifting; sport-specific modification plan

Mobility: Hip flexor/hamstring; thoracic extension (not lumbar hyperextension)

Motor Control: Hip hinge mastery; lumbopelvic dissociation

Strengthening: Bridge, hip thrust, RDL light, clamshell

Core Stability: Progress bird dog load; anti-rotation

Functional Training: Sport-specific non-extension drills

Return to Work: Gradual loading

Return to Sport: Phase 1 sport drills without extension

Progression: Strength benchmarks → functional phase

Evidence: Core stabilization reduces LBP recurrence athletes

References: McGill; APTA

RECORD: Spondylolysis / Spondylolisthesis — Remodeling/Functional

ID: REHAB-LUM-011

Condition: Spondylolysis / Spondylolisthesis

Phase: Remodeling/Functional

Goals: Full lumbopelvic control; sport-specific return

Pain Management: Extension monitoring in sport

Education: Maintenance program; technique coaching

Mobility: Full hip ROM; controlled lumbar ROM

Motor Control: Automatic hip hinge under load

Strengthening: Deadlift progression; single-leg work

Core Stability: Loaded carries; sport integration

Functional Training: Sport-specific progressions with technique modification

Return to Work: Full duty

Return to Sport: Gradual return extension activities with monitoring

Progression: Full sport participation

Evidence: Structured return reduces re-injury

References: Sports physio return-to-sport frameworks

RECORD: Spondylolysis / Spondylolisthesis — RTS/RTW

ID: REHAB-LUM-012

Condition: Spondylolysis / Spondylolisthesis

Phase: RTS/RTW

Goals: Unrestricted activity with maintenance program

Pain Management: Independent management

Education: Lifetime core maintenance for athletes

Mobility: Regular hip and thoracic mobility

Motor Control: Technique maintenance coaching

Strengthening: Periodized strength program

Core Stability: Integrated training

Functional Training: Full competition

Return to Work: Full

Return to Sport: Full with technique optimization

Progression: Maintenance

Evidence: Long-term athlete management pars defects

References: Cricket/baseball pars literature

RECORD: Facet Syndrome — Acute

ID: REHAB-LUM-013

Condition: Facet Syndrome

Phase: Acute

Goals: Reduce extension-provoked pain; restore basic ADLs

Pain Management: NSAIDs; avoid extension loading; heat

Education: Facet pain mechanism; extension intolerance common

Mobility: Flexion-based gentle ROM; knee-to-chest

Motor Control: Posterior pelvic tilt awareness

Strengthening: Minimal — isometric glute

Core Stability: Curl-up flexion bias

Functional Training: Avoid prolonged standing

Return to Work: Modified standing/sitting

Return to Sport: Pause extension sports

Progression: Extension pain reducing

Evidence: Manual therapy + exercise for LBP

References: APTA LBP CPG

RECORD: Facet Syndrome — Subacute

ID: REHAB-LUM-014

Condition: Facet Syndrome

Phase: Subacute

Goals: Flexion ROM full; begin hip and core strengthening

Pain Management: Gradual extension exposure if tolerated

Education: Posture variability; avoid fear-avoidance

Mobility: Hip flexor stretch; thoracic mobility

Motor Control: Neutral spine with hip hinge

Strengthening: Bridge, squat, row

Core Stability: McGill Big 3

Functional Training: Light functional tasks

Return to Work: Gradual standing tolerance

Return to Sport: Low-impact aerobic

Progression: Functional loading

Evidence: Exercise effective facet-related LBP subgroup

References: Cochrane manual therapy LBP

RECORD: Facet Syndrome — Remodeling/Functional

ID: REHAB-LUM-015

Condition: Facet Syndrome

Phase: Remodeling/Functional

Goals: Full function; extension tolerance if possible

Pain Management: Self-management

Education: Maintenance exercises

Mobility: Full ROM all planes per tolerance

Motor Control: Dynamic control

Strengthening: Full strength program

Core Stability: Anti-rotation and carries
Functional Training: Work and sport simulation
Return to Work: Full
Return to Sport: Extension sports with monitoring
Progression: Clearance testing
Evidence: Multimodal rehab facet syndrome
References: Bogduk facet pain management

RECORD: Facet Syndrome — RTS/RTW

ID: REHAB-LUM-016
Condition: Facet Syndrome
Phase: RTS/RTW
Goals: Independent maintenance
Pain Management: Flare plan
Education: Home program
Mobility: Maintenance
Motor Control: Automatic
Strengthening: 2-3× week maintenance
Core Stability: Ongoing
Functional Training: Full activity
Return to Work: Full
Return to Sport: Full
Progression: Maintenance
Evidence: Active lifestyle reduces chronic LBP
References: NICE NG59

RECORD: Radiculopathy — Acute

ID: REHAB-LUM-017
Condition: Radiculopathy
Phase: Acute
Goals: Centralize symptoms if discogenic; monitor neuro status; avoid cauda equina miss
Pain Management: NSAIDs/neuropathic agents per MD; positional relief
Education: Red flags; prognosis; stay active within tolerance
Mobility: Directional preference; gentle neural glides if not worsening
Motor Control: Minimal loading; protect nerve
Strengthening: Avoid if increasing leg pain
Core Stability: Gentle TA activation supine
Functional Training: Walking short distances
Return to Work: Modified; no prolonged sitting if flexion intolerant
Return to Sport: Hold
Progression: Leg pain reducing or centralizing
Evidence: Conservative care first unless cauda equina/progressive deficit
References: NICE NG59; SPORT trial

RECORD: Radiculopathy — Subacute

ID: REHAB-LUM-018

Condition: Radiculopathy

Phase: Subacute

Goals: Normalize gait; begin LE strengthening; neural mobility

Pain Management: Graded exposure

Education: Neuro recovery timeline; activity pacing

Mobility: Neural flossing/sliders per established protocols carefully

Motor Control: Bird dog; avoid SLR stretch early if provocative

Strengthening: Quad, glute, calf per affected myotome

Core Stability: McGill progressions

Functional Training: Stairs, sit-stand

Return to Work: Gradual

Return to Sport: Walking/bike

Progression: Motor strength improving

Evidence: Neural mobilization adjunct evidence moderate

References: Magee neurodynamic testing

RECORD: Radiculopathy — Remodeling/Functional

ID: REHAB-LUM-019

Condition: Radiculopathy

Phase: Remodeling/Functional

Goals: Full LE strength; reflex may lag; functional symmetry

Pain Management: Intermittent symptoms acceptable if improving trend

Education: Self-neural mobilization; maintenance

Mobility: Full neural and joint mobility

Motor Control: Dynamic single-leg stability

Strengthening: Full LE and core program

Core Stability: Loaded stabilization

Functional Training: Sport/work specific

Return to Work: Full if strength adequate

Return to Sport: Gradual run/jump if S1/L5 affected

Progression: Functional testing

Evidence: Exercise improves radiculopathy outcomes conservative group

References: APTA; Cochrane

RECORD: Radiculopathy — RTS/RTW

ID: REHAB-LUM-020

Condition: Radiculopathy

Phase: RTS/RTW

Goals: Full return; monitor neuro annually if residual numbness

Pain Management: Independent

Education: Maintenance program

Mobility: Ongoing

Motor Control: Sport automatic
Strengthening: Maintenance
Core Stability: Integrated
Functional Training: Full
Return to Work: Full
Return to Sport: Full when neuro stable and strength symmetric
Progression: Maintenance
Evidence: Most radiculopathy resolves or stabilizes without surgery
References: NICE NG59

RECORD: Nonspecific Low Back Pain — Acute

ID: REHAB-LUM-021

Condition: Nonspecific Low Back Pain

Phase: Acute

Goals: Reduce pain sufficient for ADLs; reassurance; screen red flags and yellow flags

Pain Management: Heat, NSAIDs short course, stay active — AVOID bed rest >1-2 days

Education: Favorable prognosis acute LBP; pain ≠ damage; psychosocial yellow flags (fear, catastrophizing, work dissatisfaction)

Mobility: Gentle walking; movement within tolerance; no forced stretching acute

Motor Control: Breathing; gentle pelvic tilt

Strengthening: Not primary acute unless mild

Core Stability: Gentle activation only

Functional Training: Maintain ADLs modified

Return to Work: Stay at work or early return modified preferred (NICE)

Return to Sport: Modified or pause if painful

Progression: Pain improving 1-2 weeks → subacute

Evidence: Strong: stay active, reassurance, avoid bed rest (NICE NG59, Cochrane)

References: NICE NG59; APTA LBP CPG; Cochrane acute LBP

RECORD: Nonspecific Low Back Pain — Subacute

ID: REHAB-LUM-022

Condition: Nonspecific Low Back Pain

Phase: Subacute

Goals: Restore full ROM; begin graded exercise; address yellow flags

Pain Management: Graded activity exposure; CBT/behavioral if yellow flags

Education: Self-management; flare plan; workplace ergonomics

Mobility: Hip flexor, hamstring, thoracic mobility as needed — not excessive

Motor Control: TA bracing; bird dog; dead bug

Strengthening: Bridge, squat, row, walking program

Core Stability: McGill Big 3; Pilates-style optional

Functional Training: Lifting technique introduction

Return to Work: Full or graded return

Return to Sport: Gradual aerobic

Progression: Pain controlled with daily activity → remodeling

Evidence: Exercise therapy moderate effect subacute/chronic LBP (Cochrane)

References: Cochrane exercise chronic LBP; APTA CPG

RECORD: Nonspecific Low Back Pain — Remodeling/Functional

ID: REHAB-LUM-023

Condition: Nonspecific Low Back Pain

Phase: Remodeling/Functional

Goals: Full strength and function; prevent recurrence; address psychosocial factors

Pain Management: Self-efficacy focus; pain coping strategies

Education: Independent program; recognize recurrence early

Mobility: Maintain hip/thoracic mobility

Motor Control: Automatic stabilization

Strengthening: Progressive resistance full body

Core Stability: Integrated functional training

Functional Training: Work and sport specific

Return to Work: Full duties

Return to Sport: Full participation

Progression: Functional tests passed

Evidence: Combined exercise + psychosocial best chronic LBP (guidelines)

References: NICE NG59; STaRT Back stratified care research

RECORD: Nonspecific Low Back Pain — RTS/RTW

ID: REHAB-LUM-024

Condition: Nonspecific Low Back Pain

Phase: RTS/RTW

Goals: Sustained recovery; reduced recurrence; self-management

Pain Management: Independent flare management

Education: Maintenance 1-2× week; physical activity habit

Mobility: Regular mobility practice

Motor Control: Habitual good movement patterns

Strengthening: Ongoing resistance training general health

Core Stability: Maintenance

Functional Training: Full life/work/sport participation

Return to Work: Sustained full duty

Return to Sport: Full

Progression: Long-term active lifestyle

Evidence: Physical activity reduces LBP recurrence

References: NICE NG59; WHO physical activity guidelines

12. Special Tests

RECORD: Straight Leg Raise (SLR)

ID: TEST-LUM-001

Test Name: Straight Leg Raise (Lasègue)

Purpose: Neural tension / L4-S1 radiculopathy screening

Technique: Supine passive hip flexion with knee extended; note angle of leg pain vs hamstring stretch

Positive Finding: Radicular leg pain below knee reproduced (not hamstring pull only)

Sensitivity: ~91% approx for disc herniation radiculopathy (variable by study)

Specificity: ~26% approx — low; crossed SLR increases specificity (~88% approx)

References: Magee 2014; Deville SLR systematic review

RECORD: Slump Test

ID: TEST-LUM-002

Test Name: Slump Test

Purpose: Neural tension sciatic/femoral bias

Technique: Seated slumped flexion, knee extension, ankle dorsiflexion sequential

Positive Finding: Reproduction radicular symptoms with cervical flexion component

Sensitivity: Moderate for radiculopathy

Specificity: Low-moderate

References: Magee 2014; Butler neurodynamics

RECORD: Prone Instability Test

ID: TEST-LUM-003

Test Name: Prone Instability Test

Purpose: Clinical lumbar instability subgroup

Technique: Prone, feet on floor, lift LEs — pain central segment; then ask patient brace abs and repeat

Positive Finding: Pain decreases with abdominals braced (suggests instability subgroup)

Sensitivity: ~71% approx

Specificity: ~57% approx

References: Hicks instability subgroup research; Magee 2014

RECORD: One-Legged Hyperextension (Stork Test)

ID: TEST-LUM-004

Test Name: Stork Test (One-Legged Hyperextension)

Purpose: Spondylolysis screening adolescent

Technique: Stand on one leg, extend spine; compare sides

Positive Finding: Unilateral extension pain at pars region

Sensitivity: Low (~50-65% approx); MRI/CT gold standard

Specificity: Moderate

References: Magee 2014 adolescent athlete screening

RECORD: Femoral Nerve Tension Test

ID: TEST-LUM-005

Test Name: Femoral Nerve Tension Test (Prone Knee Bend)

Purpose: L2-L4 radiculopathy / femoral nerve irritation

Technique: Prone, flex knee toward buttock; optionally extend hip

Positive Finding: Anterior thigh radicular pain (L2-L4 dermatomes)

Sensitivity: Moderate for upper lumbar radiculopathy

Specificity: Moderate

References: Magee 2014

RECORD: Crossed Straight Leg Raise

ID: TEST-LUM-006

Test Name: Crossed Straight Leg Raise (Crossed SLR)

Purpose: Sciatica / disc herniation specificity

Technique: Passive SLR symptomatic leg reproduces contralateral leg radicular pain

Positive Finding: Radicular pain in opposite leg when raising affected leg

Sensitivity: ~29% approx

Specificity: ~88% approx

References: Deville SLR systematic review; Magee 2014

RECORD: Kemp Test (Extension-Rotation)

ID: TEST-LUM-007

Test Name: Kemp Test (Quadrant Test / Extension-Rotation)

Purpose: Facet joint pain, stenosis provocation, disc (non-specific)

Technique: Seated or standing extension and rotation to same side with overpressure

Positive Finding: Reproduction familiar back/leg pain

Sensitivity: Low-moderate facet syndrome

Specificity: Low

References: Magee 2014; Bogduk facet pain

RECORD: SI Joint Cluster (Laslett)

ID: TEST-LUM-008

Test Name: Sacroiliac Joint Pain Provocation Cluster

Purpose: SI joint pain identification

Technique: Distraction, thigh thrust, FABER/Patrick, compression, Gaenslen — count positives

Positive Finding: ≥3 of 5 tests positive suggests SI joint pain (research cluster)

Sensitivity: ~94% approx cluster

Specificity: ~78% approx cluster

References: Laslett SIJ cluster research; Magee 2014

RECORD: Waddell Signs

ID: TEST-LUM-009

Test Name: Waddell Non-Organic Signs

Purpose: Behavioral/yellow flag screening in chronic LBP — NOT malingering test

Technique: Tenderness simulation, axial loading, rotation distraction, SLR distraction, regional sensory/motor

Positive Finding: ≥3 categories positive — psychosocial overlay; guide biopsychosocial approach

Sensitivity: N/A screening tool

Specificity: N/A

References: Waddell et al.; STarT Back; use cautiously without stigmatizing

RECORD: Bicycle Test (Stenosis)

ID: TEST-LUM-010

Test Name: Bicycle / Treadmill Test for Stenosis

Purpose: Differentiate neurogenic vs vascular claudication

Technique: Upright treadmill walking vs recumbent cycling — stenosis often better cycling (flexion)

Positive Finding: Walking limited, cycling tolerated suggests stenosis component

Sensitivity: Clinical observation

Specificity: Moderate vs vascular

References: Neurogenic claudication clinical exam; Magee 2014

13. Clinical Examination Template and Red Flag Screen

RECORD: Lumbar Spine Clinical Examination

Template Name: Lumbar Spine Clinical Examination

ID: EXAM-LUM-001

Subjective: Onset, mechanism, location (axial vs leg), dermatomal symptoms, aggravating/easing, bowel/bladder, saddle symptoms, fever, weight loss, cancer history, steroid use, night pain

Red Flag Screen: Cauda equina (saddle, retention, bilateral weakness), fracture (trauma, osteoporosis), infection (fever, IV drug use), malignancy (age >50 unexplained, history, weight loss), progressive deficit

Yellow Flags: Catastrophizing, fear-avoidance, depression, work dissatisfaction, litigation — STarT Back tool

Observation: Posture, antalgic list, gait, scoliosis, muscle wasting

ROM: Flexion finger-to-floor, extension, lateral flexion, rotation; repeated movements McKenzie

Neurological: Dermatomes L1-S1, myotomes (EHL L5, GSC S1, TA, quads), reflexes patellar L4 Achilles S1, SLR, crossed SLR, slump

Special Tests: SI cluster, Kemp extension-rotation, stenosis walking test, Waddell signs if chronic

Palpation: Spinous processes, facets, paraspinal, QL, SI region

Functional: Squat, single-leg stance, hip ROM, neurodynamic tests

Imaging Indications: Red flags, progressive neuro deficit, cauda equina emergency MRI, no improvement 6 weeks with radiculopathy consider MRI per NICE

References: Magee 2014; NICE NG59; STarT Back tool

RECORD: Lumbar Red Flag Screening (Emergency Referral)

Template Name: Lumbar Red Flag Screening (Emergency Referral)

ID: RF-LUM-001

Cauda Equina: URGENT: urinary retention, overflow incontinence, fecal incontinence, saddle anesthesia, bilateral sciatica, progressive bilateral weakness, reduced anal sphincter tone — EMERGENCY MRI and surgical consult

Fracture: Major trauma, minor trauma osteoporosis/steroids, prolonged corticosteroid use, age >70 with trauma

Malignancy: Age >50 new unexplained LBP, history cancer, unexplained weight loss, night pain unrelieved by rest

Infection: Fever, IV drug use, recent spinal procedure, immunosuppression, severe constant pain

Inflammatory: Morning stiffness >30 min improving with activity, alternating buttock pain (consider AS), family history

Progressive Neurological Deficit: Worsening motor weakness, new foot drop, ascending sensory loss — urgent imaging

Action: Do not continue conservative trial as primary management when cauda equina or progressive deficit suspected; direct emergency department or spinal on-call referral; document time of referral

References: NICE NG59 red flags; APTA LBP CPG; Magee 2014

14. Exercise Library

RECORD: McGill Curl-Up

ID: EX-LUM-001

Exercise: McGill Curl-Up

Target: Rectus abdominis, transversus abdominis — minimal lumbar flexion

Phase: Acute-subacute LBP, disc herniation, nonspecific LBP

Technique: Supine, one knee bent, hands under lumbar to maintain neutral; lift head/shoulders slightly without low back flattening

Dosage: 3 sets × 10-15 reps; hold 8-10 sec optional isometric variant

Cues: Brace before lift; avoid pulling on neck; breathe normally

Contraindications: Acute severe flexion intolerance (modify range); cauda equina — not applicable (emergency)

Progression: Increase hold time → feet on floor both bent → add arm reach

Evidence: Spine-sparing abdominal training (McGill)

References: McGill Low Back Disorders

RECORD: Bird Dog (Quadruped Opposite Arm/Leg)

ID: EX-LUM-002

Exercise: Bird Dog

Target: Multifidus, erector spinae, gluteus maximus, transversus abdominis

Phase: Subacute-remodeling all LBP conditions except acute severe pain

Technique: Quadruped neutral spine; extend opposite arm/leg to horizontal without trunk rotation or lumbar sag

Dosage: 3 sets × 8-10 each side; 5-10 sec holds

Cues: Keep pelvis level; brace core; slow controlled return

Contraindications: Wrist pain (modify on bench); acute extension intolerance

Progression: Add holds → draw squares with foot → light cuff weight

Evidence: Core stability staple; moderate EMG multifidus/glute

References: McGill Big 3; APTA LBP CPG

RECORD: Side Plank

ID: EX-LUM-003

Exercise: Side Plank

Target: Quadratus lumborum, obliques, gluteus medius

Phase: Subacute onward

Technique: Forearm or hand support; body straight line; hips elevated; knees bent (modified) or legs straight

Dosage: 3 sets × 20-45 sec each side

Cues: Don't let hips drop; breathe; top arm rest on hip or reach ceiling

Contraindications: Acute shoulder injury; acute lateral pain that worsens

Progression: Knees bent → full plank → add top leg abduction

Evidence: McGill Big 3; high QL EMG

References: McGill

RECORD: Bridge / Hip Thrust

ID: EX-LUM-004

Exercise: Bridge / Hip Thrust

Target: Gluteus maximus, hamstrings, core stabilization

Phase: Subacute-remodeling; essential disc/radiculopathy/spondylolisthesis

Technique: Supine feet flat hip-width; posterior pelvic tilt; lift hips to neutral alignment without hyperextension

Dosage: 3 sets × 12-15; progress to barbell hip thrust

Cues: Drive through heels; squeeze glutes top; don't over-arch lumbar

Contraindications: Acute extension intolerance (limit height)

Progression: Double leg → single leg → weighted hip thrust

Evidence: Hip extension offloads lumbar erectors in lifting models

References: McGill; glute activation LBP literature

RECORD: Dead Bug

ID: EX-LUM-005

Exercise: Dead Bug

Target: Transversus abdominis, obliques, hip flexors with lumbar stability

Phase: Subacute all conditions

Technique: Supine 90/90 hips/knees; lower opposite arm/leg while maintaining neutral lumbar (press back to floor lightly)

Dosage: 3 sets × 8-10 each side slow

Cues: Exhale on extension; small range if losing neutral

Contraindications: Acute severe pain with leg lowering

Progression: Band resistance → straight leg lower

Evidence: Motor control training moderate evidence

References: Hodges/Richardson; APTA CPG

RECORD: Prone Press-Up (Extension in Lying)

ID: EX-LUM-006

Exercise: Prone Press-Up (McKenzie Extension)

Target: Lumbar extension ROM; centralization disc herniation

Phase: Acute-subacute disc herniation with extension preference ONLY

Technique: Prone, hands under shoulders; press upper body up keeping pelvis down; oscillate or hold

Dosage: 10-15 reps × 3-4 sessions/day if centralizing per McKenzie

Cues: Monitor symptom response — must centralize or improve; stop if peripheralization

Contraindications: Stenosis, spondylolysis acute, cauda equina, extension worsens symptoms

Progression: Add overpressure → standing extension

Evidence: McKenzie mechanical diagnosis; extension preference subgroup benefit

References: McKenzie Treat Your Own Back; clinical reasoning

RECORD: Hip Hinge / Romanian Deadlift Pattern

ID: EX-LUM-007

Exercise: Hip Hinge / Romanian Deadlift (RDL)

Target: Gluteus maximus, hamstrings, erector spinae isometric

Phase: Remodeling-functional; RTW/RTS

Technique: Soft knees, push hips back, neutral spine, bar/DB close to legs; feel hamstring stretch not lumbar flexion

Dosage: 3 sets × 8-12 progressive load

Cues: Chest up; lats engaged; bar path vertical over mid-foot

Contraindications: Acute radiculopathy uncontrolled; flexion intolerance without modification

Progression: PVC pipe → dumbbell → barbell → single-leg RDL

Evidence: Functional lifting retraining cornerstone

References: McGill; NIOSH lifting; APTA RTW

RECORD: Pallof Press (Anti-Rotation)

ID: EX-LUM-008

Exercise: Pallof Press

Target: Obliques, transversus abdominis, multifidus anti-rotation

Phase: Subacute-remodeling

Technique: Cable/band at chest height; press forward resisting rotation; hold 3-5 sec

Dosage: 3 sets × 10-12 each side

Cues: Don't rotate trunk; hips square; breathe

Contraindications: Acute rotation pain

Progression: Increase band tension → half-kneeling → split stance

Evidence: Anti-rotation core training functional carryover

References: APTA core stability programs

RECORD: Stationary Bike (Flexion Bias)

ID: EX-LUM-009

Exercise: Stationary Cycling

Target: Aerobic conditioning; hip/knee ROM; flexion-biased stenosis

Phase: Subacute stenosis, chronic LBP, all conditions needing low-impact cardio

Technique: Seat height hip flexion ~30 deg; upright or slight forward lean for stenosis

Dosage: 20-30 min moderate intensity; progress duration before intensity

Cues: Neutral spine; avoid slumped flexion if disc intolerant

Contraindications: Severe flexion intolerance (adjust seat)

Progression: Duration → interval → outdoor cycling

Evidence: Aerobic exercise recommended NICE NG59; stenosis flexion relief

References: NICE NG59; NASS stenosis

RECORD: Cat-Camel (Spinal Mobility)

ID: EX-LUM-010

Exercise: Cat-Camel

Target: Lumbar/thoracic mobility; disc nutrition motion

Phase: Subacute nonspecific LBP; warm-up

Technique: Quadruped slow flexion-extension cycles through available ROM without forcing end range

Dosage: 10-15 slow cycles × 2-3 sets

Cues: Move segmentally if possible; pain-free range

Contraindications: Acute severe pain either direction; stenosis extension end-range

Progression: Add rotation in quadruped (thread the needle)

Evidence: Gentle mobility adjunct; not standalone treatment

References: APTA LBP CPG

RECORD: Neural Glide — Sciatic (Slump Slider)

ID: EX-LUM-011

Exercise: Sciatic Neural Glide (Slump Slider)

Target: Sciatic nerve mobility; radiculopathy L4-S1

Phase: Subacute radiculopathy when irritability decreasing

Technique: Seated slump, extend knee then dorsiflex; flex knee to slacken; oscillate

Dosage: 10-15 slow reps × 2-3 sets; monitor 24h response

Cues: Do not push into severe pain; differentiate hamstring vs neural

Contraindications: High irritability radiculopathy; cauda equina; progressive deficit

Progression: Add cervical flexion component carefully

Evidence: Adjunct neural mobilization moderate evidence

References: Butler/Mobilisation of the Nervous System; Magee

RECORD: Clamshell

ID: EX-LUM-012

Exercise: Clamshell

Target: Gluteus medius, deep hip external rotators

Phase: Subacute all LBP — lumbopelvic stability

Technique: Side-lying hips/knees flexed 45 deg; open top knee without pelvis rolling back

Dosage: 3 sets × 15-20 each side; band above knees progress

Cues: Heels together; pelvis stacked; controlled tempo

Contraindications: Hip pathology limiting side-lying

Progression: Band resistance → side-lying hip abduction

Evidence: Glute med weakness associated with LBP in some studies

References: Hip-spine syndrome literature

RECORD: Suitcase Carry

ID: EX-LUM-013

Exercise: Suitcase Carry (Unilateral Farmer Carry)

Target: QL, obliques, gluteus medius, grip, functional anti-lateral flexion

Phase: Remodeling-functional; RTW

Technique: Single dumbbell/kettlebell at side; walk tall without lateral lean; neutral spine

Dosage: 3 sets × 20-40 m each hand

Cues: Don't shrug toward weight; brace core; even steps

Contraindications: Acute lateral pain; severe stenosis walking limited

Progression: Heavier load → longer distance → overhead carry variant

Evidence: Functional core training; occupational carry simulation

References: McGill carries; APTA functional training

RECORD: Walking Program (Graded)

ID: EX-LUM-014

Exercise: Graded Walking Program

Target: Aerobic fitness, disc nutrition, general LBP recovery
Phase: Acute onward — first-line activity NICE
Technique: Start pain-free or mildly acceptable distance; increase 10-15% per week; upright posture
Dosage: Daily or 5×/week; 20-45 min goal
Cues: Arm swing natural; avoid shopping cart unless stenosis (then flexion OK)
Contraindications: Neurogenic claudication — use pacing and flexion breaks
Progression: Distance → speed → incline → weighted vest optional
Evidence: Strong stay-active recommendation NICE NG59
References: NICE NG59; Cochrane walking LBP

RECORD: Pool Therapy / Aquatic Exercise

ID: EX-LUM-015
Exercise: Aquatic Therapy
Target: De-loaded spinal movement; aerobic and strengthening
Phase: Acute-subacute when land-based exercise intolerant; stenosis, spondylolysis
Technique: Walking in pool, water aerobics, gentle swimming; buoyancy reduces axial load
Dosage: 30-45 min sessions 2-3×/week
Cues: Maintain neutral spine with flutter board if swimming
Contraindications: Open wounds; infection risk; fear of water
Progression: Reduce water depth → land-based transition
Evidence: Moderate benefit chronic LBP aquatic exercise reviews
References: Cochrane aquatic exercise LBP

RECORD: Thoracic Rotation Mobility (Open Book)

ID: EX-LUM-016
Exercise: Open Book / Thread the Needle
Target: Thoracic rotation mobility — reduces compensatory lumbar rotation
Phase: Subacute onward
Technique: Side-lying or quadruped rotation with thoracic emphasis; block lumbar if needed
Dosage: 10 reps each side × 2 sets daily
Cues: Rotate through mid-back not low back
Contraindications: Acute rotation-provoked pain
Progression: Add foam roller thoracic extension
Evidence: Thoracic hypomobility linked to LBP compensations
References: Manual therapy LBP adjunct evidence

RECORD: Hip Flexor Stretch (Half-Kneeling)

ID: EX-LUM-017
Exercise: Half-Kneeling Hip Flexor Stretch
Target: Iliopsoas, rectus femoris
Phase: Subacute all LBP with hip flexor tightness
Technique: Half-kneeling, posterior pelvic tilt, lean forward until anterior hip stretch felt without lumbar extension
Dosage: 3 sets × 30-45 sec each side
Cues: Tuck pelvis first; don't arch low back

Contraindications: Knee pain on floor (pad); hip pathology

Progression: Overhead reach same side → add light contract-relax

Evidence: Hip flexor length part of lumbopelvic management

References: Magee 2014 hip examination

RECORD: Wall Squat / Sit-to-Stand

ID: EX-LUM-018

Exercise: Wall Squat / Sit-to-Stand

Target: Quadriceps, glutes, functional ADL

Phase: Subacute-remodeling; stenosis (flexion OK)

Technique: Back against wall slide to comfortable depth; or chair sit-to-stand without arms

Dosage: 3 sets × 10-15

Cues: Knees track toes; chest up; hip hinge on stand

Contraindications: Patellofemoral pain modify depth

Progression: Single leg sit-to-stand → weighted squat

Evidence: Functional strengthening NICE recommended exercise

References: NICE NG59; APTA

15. Imaging Recommendations

RECORD: Lumbar X-Ray (AP/Lateral)

ID: IMG-LUM-001

Modality: Lumbar X-Ray AP and Lateral

Indications: Trauma, suspected fracture, spondylolisthesis grading, scoliosis screening, pre-operative baseline, failed conservative care needing alignment assessment

Views: AP, lateral standing (flexion-extension if instability suspected)

Findings: Lordosis angle, disc height loss, osteophytes, pars defect (oblique views), Meyerding grade listhesis, spondylolisthesis, sacralization L5

Limitations: Poor soft tissue/disc/nerve visualization; radiation exposure

Red Flags: Fracture line, listhesis progression, destructive lesion

References: ACR Appropriateness; Magee 2014

RECORD: Lumbar MRI

ID: IMG-LUM-002

Modality: Lumbar MRI (without contrast first line)

Indications: Red flags, cauda equina EMERGENCY, progressive neuro deficit, persistent radiculopathy >6 weeks, suspected infection/malignancy (with contrast), pre-surgical planning

Findings: Disc herniation type (bulge/protrusion/extrusion/sequestration), nerve root compression, central/lateral recess/foraminal stenosis, Modic changes, Pfirrmann grade, facet effusion, cord/conus level

Limitations: High false-positive disc findings asymptomatic adults; correlate clinically

Red Flags: Cauda equina compression, epidural abscess, metastatic lesion, conus lesion

References: NICE NG59 imaging; ACR Appropriateness; Jensen NEJM incidental disc findings

RECORD: Lumbar CT

ID: IMG-LUM-003

Modality: Lumbar CT (often with myelography legacy; CT myelogram selective)

Indications: Pars defect characterization, post-operative fusion assessment, MRI contraindicated (pacemaker rare now), fracture detail, surgical planning bony anatomy

Findings: Pars lysis, facet tropism, bony stenosis dimensions, fusion mass

Limitations: Radiation; limited direct nerve visualization without myelogram

References: Magee 2014; adolescent spondylolysis protocols

RECORD: Bone Scan (SPECT-CT)

ID: IMG-LUM-004

Modality: Nuclear Medicine Bone Scan / SPECT-CT

Indications: Suspected metastases, occult fracture, active pars stress (SPECT), infection, inflammatory conditions

Findings: Increased uptake active pars, metastatic lesions, facet inflammation

Limitations: Non-specific; MRI often preferred

References: Oncology red flag workup; sports pars imaging

RECORD: EMG/NCS Lumbar

ID: IMG-LUM-005

Modality: Electromyography and Nerve Conduction Studies

Indications: Subacute/chronic radiculopathy confirmation, differentiate peripheral neuropathy, pre-surgical baseline, multilevel disease

Findings: Denervation potentials affected myotome, fibrillations, chronic reinnervation

Limitations: May be normal early acute radiculopathy; does not show compressive level directly always

References: AANEM guidelines; Magee 2014

16. Return to Sport / Return to Work Criteria

RECORD: Disc Herniation — Return to Sport

ID: RTS-LUM-001

Condition: Disc Herniation

Criteria: Leg pain resolved or minimal; full ROM; SLR negative; strength $\geq 90\%$ bilateral LE; no neurological deficit; sport-specific drills without peripheralization

Testing: Single-leg hop symmetry, sport-specific simulation 24h symptom check

Timeline: Variable — weeks to months; no fixed calendar — criteria-based

References: Sports medicine return-to-play consensus

RECORD: Spondylolysis — Return to Sport

ID: RTS-LUM-002

Condition: Spondylolysis / Spondylolisthesis

Criteria: Pain-free extension-rotation sport movements; CT/MRI stable pars; core and hip strength symmetric; MD clearance adolescent athletes

Testing: Sport-specific throwing/bowling/gymnastics progressions monitored

Timeline: 3-6 months minimum often; high-load sports longer

References: Adolescent athlete pars return protocols

RECORD: Lumbar Stenosis — Return to Activity

ID: RTS-LUM-003

Condition: Lumbar Stenosis

Criteria: Walking tolerance meets functional goals (often modified); flexion strategies mastered; no progressive neuro deficit

Testing: Timed walk test; community ambulation distance

Timeline: Chronic management — ongoing pacing

References: NASS stenosis functional outcomes

RECORD: Nonspecific LBP — Return to Work

ID: RTS-LUM-004

Condition: Nonspecific Low Back Pain

Criteria: Vocational tasks simulated pain-free; lifting capacity meets job demands; yellow flags addressed

Testing: Material handling assessment; sit-stand tolerance; job-specific simulation

Timeline: Early return modified work preferred NICE — often days to 2 weeks

References: NICE NG59 return to work; occupational health guidelines

RECORD: Radiculopathy — Return to Sport

ID: RTS-LUM-005

Condition: Radiculopathy

Criteria: Myotome strength full; reflex may still be diminished if stable; no progressive sensory loss; running/jumping without leg pain

Testing: Neurological re-exam; functional LE tests; gradual sport progression

Timeline: 6-12 weeks typical uncomplicated; longer if surgical

References: APTA; SPORT trial follow-up concepts

17. Outcome Measures

RECORD: Oswestry Disability Index (ODI)

ID: OM-LUM-001

Measure: Oswestry Disability Index (ODI)

Type: Patient-reported outcome — functional disability

Items: 10 sections: pain intensity, personal care, lifting, walking, sitting, standing, sleeping, sex life, social life, traveling

Scoring: 0-100%; minimal disability 0-20%; moderate 21-40%; severe 41-60%; crippled 61-80%; bed-bound 81-100%

MCID: Minimum clinically important difference ~10-15 points (approximate; varies by study)

Use: Track LBP disability over treatment; research standard

References: Fairbank ODI; orthopedic outcome measure literature

RECORD: Numeric Pain Rating Scale (NPRS)

ID: OM-LUM-002

Measure: Numeric Pain Rating Scale (NPRS 0-10)

Type: Pain intensity patient-reported

Items: Single item current pain 0 no pain to 10 worst imaginable

Scoring: 0-10 continuous

MCID: ~2 points approx for LBP (variable)

Use: Quick serial pain monitoring; telehealth friendly

References: IMMPACT recommendations chronic pain

RECORD: STarT Back Screening Tool

ID: OM-LUM-003

Measure: STarT Back Screening Tool

Type: Risk stratification — psychosocial yellow flags

Items: 9 items: 5 subgroups psychosocial distress; 4 physical; subscore for high-risk

Scoring: Low, medium, high risk subgroups guide matched treatment pathways

Use: Primary care and PT triage — high-risk may benefit targeted psychologically informed PT

References: Hill STarT Back trial; UK stratified care model

RECORD: Quebec Back Pain Disability Scale

ID: OM-LUM-004

Measure: Quebec Back Pain Disability Scale

Type: Functional disability scale

Items: 20 items daily activities

Scoring: 0-100 disability score

Use: Alternative to ODI; validated LBP

References: Quebec scale validation studies

RECORD: Patient-Specific Functional Scale (PSFS)

ID: OM-LUM-005

Measure: Patient-Specific Functional Scale (PSFS)

Type: Individualized functional outcome

Items: Patient identifies up to 3 important activities rated 0-10 ability

Scoring: Track patient-nominated activities

MCID: ~2 points approx per activity

Use: Goal-oriented rehab; patient-centered outcomes

References: Stratford PSFS validation

RECORD: Fear-Avoidance Beliefs Questionnaire (FABQ)

ID: OM-LUM-006

Measure: Fear-Avoidance Beliefs Questionnaire (FABQ)

Type: Yellow flag psychosocial measure

Items: FABQ-Work and FABQ-Physical activity subscales

Scoring: Higher scores = greater fear-avoidance beliefs

Use: Identify patients needing cognitive-behavioral approaches; work-related fear

References: Waddell FABQ; chronic LBP biopsychosocial model

18. Clinical Reasoning Pathways

RECORD: Acute Low Back Pain with Leg Pain — Initial Pathway

ID: PATHWAY-LUM-001

Presentation: Acute LBP with leg pain (<6 weeks)

Step 1: Red flag screen FIRST — cauda equina, fracture, infection, malignancy, progressive deficit

Step 2: If red flags → emergency referral/imaging per protocol; STOP conservative primary pathway

Step 3: Neurological exam: dermatomes L1-S1, myotomes, patellar/Achilles reflexes, SLR, crossed SLR

Step 4: Classify: radiculopathy vs referred somatic pain vs SI/hip; leg pain below knee suggests radicular

Step 5: Education: favorable prognosis, stay active, avoid bed rest; analgesia per MD

Step 6: Physical therapy: directional preference assessment; neural mobilization if radicular low irritability

Step 7: Reassess 2-4 weeks; if progressive neuro deficit or cauda equina symptoms → urgent MRI

Step 8: Imaging if not improving 6 weeks persistent radiculopathy per NICE or earlier if red flags

References: NICE NG59; APTA LBP CPG; Magee examination

RECORD: Chronic Low Back Pain — Biopsychosocial Pathway

ID: PATHWAY-LUM-002

Presentation: Chronic LBP (>12 weeks)

Step 1: Red flag re-screen; STarT Back or FABQ yellow flag assessment

Step 2: ODI or PSFS baseline functional measure

Step 3: Subgroup: movement preference (flexion/extension/intolerance), instability signs, stenosis pattern, chronic widespread

Step 4: Graded exercise program — type less important than adherence (Cochrane)

Step 5: Manual therapy short course adjunct if indicated

Step 6: CBT or psychologically informed PT if high yellow flags (STarT Back high-risk)

Step 7: Work participation emphasis — modified duties vs prolonged disability

Step 8: Reassess 6-8 weeks with ODI/NPRS MCID tracking

References: NICE NG59 chronic LBP; STarT Back; Cochrane exercise

RECORD: Neurogenic Claudication — Stenosis Pathway

ID: PATHWAY-LUM-003

Presentation: Leg pain/heaviness with walking/standing, relieved by sitting/flexion

Step 1: Differentiate vascular claudication — pulses, ABI if indicated, bicycle vs walking test

Step 2: Red flags: acute cauda equina rare in isolated stenosis but screen bowel/bladder

Step 3: Confirm neurogenic pattern: shopping cart sign, flexion relief, bilateral often

Step 4: Conservative PT: flexion-biased exercises, cycling, hip flexor strengthening, pacing education

Step 5: Imaging MRI if not previously done and symptoms functionally limiting for shared decision

Step 6: Epidural injection trial selective cases

Step 7: Surgical decompression discussion if moderate-severe refractory to conservative 3-6 months

References: NASS lumbar stenosis guidelines; neurogenic claudication literature

RECORD: Adolescent Athlete Back Pain — Pars Pathway

ID: PATHWAY-LUM-004

Presentation: Adolescent athlete extension-intolerant LBP

Step 1: Red flags standard; high index pars suspicion gymnastics/cricket/football/dance
Step 2: Stork test, extension-rotation pain; reduce extension sports immediately
Step 3: Imaging: MRI with STIR for bone edema early; CT for pars defect confirmation
Step 4: Bracing and activity modification 3-6 months per sports medicine protocol
Step 5: Rehab: McGill Big 3, glute/hip focus, no end-range extension loading
Step 6: Gradual return sport-specific with technique modification; MD clearance
Step 7: Spondylolisthesis grading if bilateral pars defect — monitor slip standing X-ray
References: Magee adolescent spine; sports spondylolysis consensus

RECORD: Cauda Equina Suspicion — Emergency Pathway

ID: PATHWAY-LUM-005

Presentation: Suspected cauda equina syndrome

Step 1: IMMEDIATE — do not schedule routine PT follow-up as primary action
Step 2: History: urinary retention, overflow incontinence, saddle numbness, bilateral leg symptoms, fecal incontinence, progressive weakness
Step 3: Exam: perianal sensation, anal tone (physician), post-void residual bladder scan, bilateral neuro exam
Step 4: Emergency MRI whole lumbar spine — contact on-call spinal service/ED
Step 5: If confirmed → emergency surgical decompression; document time symptom onset (time-sensitive outcomes)
Step 6: PT role: post-operative rehabilitation ONLY after surgical management
Step 7: Medicolegal documentation critical — clear referral communication
References: NICE NG59 cauda equina; ACR MRI appropriateness emergency; Bogduk

RECORD: Return-to-Work Decision Pathway

ID: PATHWAY-LUM-006

Presentation: Occupational LBP — RTW planning

Step 1: Job demands analysis: sitting/standing/lifting/frequency
Step 2: Yellow flags: work dissatisfaction, FABQ-work elevated, litigation — address biopsychosocial
Step 3: Early RTW modified preferred over prolonged sick leave (NICE)
Step 4: Functional capacity evaluation: material handling, sit tolerance, walking distance
Step 5: Graded exposure to job tasks simulation in clinic
Step 6: Employer communication with patient consent — ergonomic modifications
Step 7: Full duty when job simulation pain-free and strength criteria met
References: NICE NG59 RTW; occupational health LBP guidelines

19. Manual Therapy Techniques

RECORD: Posterior-Anterior (PA) Central Mobilization

ID: MT-LUM-001

Technique: PA Central Vertebral Mobilization

Grade: Maitland Grade I-II pain modulation; III-IV stiffness (IV small amplitude end-range)

Indications: Segmental hypomobility, nonspecific LBP, facet hypomobility

Contraindications: Acute disc herniation with peripheralization on extension, cauda equina, fracture, malignancy, hypermobility instability

Technique Detail: Patient prone; central pressure over spinous process; oscillatory or sustained

Evidence: Manual therapy adjunct moderate short-term benefit LBP (Cochrane)

References: Maitland; APTA LBP CPG

RECORD: Flexion Distraction

ID: MT-LUM-002

Technique: Flexion Distraction (Cox technique concept)

Grade: Rhythmic flexion-distraction table or manual equivalent

Indications: Disc herniation flexion-intolerant alternative, stenosis, facet pain

Contraindications: Cauda equina, fracture, abdominal aortic aneurysm, pregnancy considerations

Technique Detail: Gentle repetitive flexion traction segment; monitor symptom centralization

Evidence: Limited high-quality RCTs; widely used clinically

References: Cox flexion distraction literature

RECORD: Muscle Energy — Pelvic Rotation

ID: MT-LUM-003

Technique: Muscle Energy for Pelvic/Lumbar Rotation Dysfunction

Grade: Isometric patient effort 3-5 sec, 3-5 reps

Indications: Rotational dysfunction, SI region, mild segmental restriction

Contraindications: Acute severe pain, hypermobility, acute disc

Technique Detail: Patient positioned in restriction barrier; isometric into rotation against therapist resistance; reassess

Evidence: Manual therapy component multimodal care

References: Magee 2014; osteopathic/PT manual therapy texts

RECORD: Neural Mobilization — Slump Technique

ID: MT-LUM-004

Technique: Slump Neural Mobilization (Treatment)

Grade: Oscillatory neural glide not sustained aggressive stretch

Indications: Radiculopathy low irritability, neural tension positive tests

Contraindications: High irritability radiculopathy, cauda equina, progressive deficit

Technique Detail: Seated slump sequence with knee/ankle components; monitor 24h symptom response

Evidence: Adjunct moderate evidence neural mobilization sciatica

References: Butler; Magee neurodynamics

RECORD: Soft Tissue — Quadratus Lumborum

ID: MT-LUM-005

Technique: QL Soft Tissue Mobilization / Trigger Point Release

Grade: Ischemic compression, myofascial release, dry needling where licensed

Indications: QL trigger point, lateral LBP, somatic hip referral

Contraindications: Renal pathology left side caution; anticoagulation; acute radiculopathy misidentified

Technique Detail: Side-lying access to QL between iliac crest and 12th rib; avoid excessive depth

Evidence: Short-term pain relief; combine with exercise for lasting effect

References: Travell & Simons; dry needling LBP reviews

20. Differential Diagnosis by Presentation

RECORD: Leg Pain Below Knee — Differential

ID: DDX-LUM-001

Chief Complaint: Leg pain below knee

Most Likely MSK: Lumbar radiculopathy L4/L5/S1 disc herniation or stenosis

Alternative MSK: SI joint referral, hip OA, hamstring strain, piriformis syndrome

Key Discriminators: Dermatomal pattern, reflex change, SLR positive, myotome weakness

References: Magee 2014

RECORD: Back Pain Only — Differential

ID: DDX-LUM-002

Chief Complaint: Axial low back pain only

Most Likely MSK: Nonspecific LBP, facet syndrome, myofascial pain, SI joint

Red Flags: Fever, cancer history, trauma, cauda equina if bladder symptoms develop

References: NICE NG59; Magee 2014

RECORD: Bilateral Leg Symptoms — Differential

ID: DDX-LUM-006

Chief Complaint: Bilateral leg symptoms

Urgent: Cauda equina — saddle anesthesia, bladder/bowel dysfunction — EMERGENCY MRI

Stenosis: Neurogenic claudication — extension worse, flexion better, gradual onset

References: NICE NG59

21. Lumbosacral Junction Reference

RECORD: Lumbosacral Junction (L5-S1)

ID: LSJ-LUM-001

Structure: Lumbosacral Junction

Anatomy: L5 on sacrum; highest shear; iliolumbar ligaments

Common Pathology: L5-S1 disc herniation; spondylolysis L5; listhesis

References: Bogduk 2012

RECORD: Conus vs Cauda Equina

ID: LSJ-LUM-003

Structure: Conus Medullaris vs Cauda Equina

Clinical Distinction: Conus: UMN, symmetric, early bladder; Cauda: LMN, asymmetric roots L2-S5

Emergency: Urgent MRI both presentations when suspected

References: Netter; Bogduk 2012

RECORD: Dermatomal Reference L1-S1

ID: LSJ-LUM-004

Structure: Dermatome/Myotome/Reflex Quick Reference

L4: Medial leg — patellar reflex — quadriceps/TA

L5: Lateral leg/dorsum foot — EHL — no primary DTR

S1: Posterior leg — Achilles reflex — plantarflexion

References: Netter; Magee 2014

22. Patient Education Topics

RECORD: Stay Active — Avoid Bed Rest

ID: EDU-LUM-001

Topic: Stay Active; Avoid Prolonged Bed Rest

Key Message: Acute LBP: continue normal activities within pain tolerance; bed rest >1-2 days not recommended

Rationale: Maintains conditioning, disc nutrition, reduces fear-avoidance; faster recovery vs rest

Practical Advice: Short walks, modify not stop work if possible, pain does not equal harm for nonspecific LBP

Cautions: Modify if radiculopathy; emergency if cauda equina symptoms

References: NICE NG59 strong recommendation

RECORD: Understanding Disc Herniation

ID: EDU-LUM-002

Topic: Patient Education — Disc Herniation

Key Message: Many disc herniations improve without surgery; leg pain may centralize to back (good sign)

Rationale: Natural history favorable for majority; MRI findings common in asymptomatic adults

Practical Advice: Avoid prolonged sitting if flexion intolerant; walking; positional relief; report bowel/bladder changes immediately

Cautions: Cauda equina emergency signs must be reviewed

References: NICE NG59 sciatica; Jensen incidental disc findings

RECORD: Red Flags — When to Seek Emergency Care

ID: EDU-LUM-003

Topic: Red Flags Patient Education

Key Message: Seek emergency care for: inability to urinate, saddle numbness, bilateral leg weakness, fecal incontinence, fever with back pain after procedure

Rationale: Cauda equina, infection, fracture may need urgent treatment

Practical Advice: Provide written red flag list; ensure patient understands bladder symptoms

References: NICE NG59; APTA LBP CPG

RECORD: Yellow Flags and Recovery

ID: EDU-LUM-004

Topic: Yellow Flags — Psychosocial Recovery Factors

Key Message: Fear of movement, belief that back is damaged, work dissatisfaction predict slower recovery — addressable

Rationale: Biopsychosocial model; STarT Back stratified care improves outcomes high-risk patients

Practical Advice: Graded exposure, realistic prognosis, stay at work modified, CBT if needed

References: STarT Back trial; FABQ literature

RECORD: Lifting and Work Ergonomics

ID: EDU-LUM-005

Topic: Lifting Education and Ergonomics

Key Message: Hip hinge, load close to body, avoid twist under load; vary posture — no single perfect posture

Rationale: Reduces flexion-compression injury risk; occupational LBP multifactorial

Practical Advice: Micro-breaks from sitting; sit-stand options; job rotation

References: NIOSH lifting; NICE RTW guidance

RECORD: Stenosis — Walking and Pacing

ID: EDU-LUM-006

Topic: Neurogenic Claudication Patient Education

Key Message: Leg heaviness when walking that eases with sitting or leaning forward (shopping cart) is common in stenosis

Rationale: Flexion opens spinal canal; pacing prevents symptom escalation

Practical Advice: Use cart in supermarket, stationary bike, flexion stretches before walking

References: NASS stenosis patient education

RECORD: Spondylolysis in Young Athletes

ID: EDU-LUM-007

Topic: Adolescent Athlete Pars Education

Key Message: Extension-sport back pain needs medical assessment; rest from extension loading allows pars healing

Rationale: Early diagnosis improves healing chance; chronic non-union may still be managed conservatively

Practical Advice: Core stability, bracing if prescribed, gradual return with technique modification

References: Sports medicine spondylolysis consensus

RECORD: Self-Management and Flare Plan

ID: EDU-LUM-008

Topic: Chronic LBP Self-Management Plan

Key Message: Flares common; return to baseline exercises not full rest; maintain aerobic activity

Rationale: Self-efficacy reduces recurrence and healthcare utilization

Practical Advice: Home exercise program 2-3× week maintenance; heat; resume walking early in flare

References: NICE NG59 chronic LBP; Cochrane self-management

23. Supplementary Clinical Reference

RECORD: L5 Transitional Vertebra — Lumbosacral Variant

ID: SUP-LUM-001

Topic: L5 Transitional Vertebra (Bertolotti Syndrome)

Definition: Partial or complete fusion L5 transverse process to sacrum/ilium; sacralization or lumbarization S1

Clinical Relevance: Altered biomechanics; may predispose to adjacent segment disease; pars stress at transition

Imaging: Castellvi classification; identify on AP X-ray enlarged transverse process

Management: Same LBP principles; address adjacent segment overload

References: Bogduk; spinal variant anatomy

RECORD: Modic Changes — Endplate MRI Classification

ID: SUP-LUM-002

Topic: Modic Endplate Changes

Type 1: Edema/inflammation — low T1, high T2; may correlate with active discogenic pain

Type 2: Fatty replacement — high T1 and T2

Type 3: Sclerosis — low T1 and T2

Clinical Relevance: Type 1 may indicate inflammatory disc pathology; weak correlation with symptoms overall

References: Modic et al.; lumbar MRI interpretation

RECORD: Pfirrmann Disc Degeneration Grades

ID: SUP-LUM-003

Topic: Pfirrmann MRI Disc Grading

Grade I: Homogeneous bright white nucleus; normal disc height

Grade II: Inhomogeneous structure; clear nucleus/annulus distinction

Grade III: Inhomogeneous gray; unclear nucleus/annulus boundary

Grade IV: Homogeneous gray; no distinction; disc height loss may begin

Grade V: Collapsed disc space; black signal

Clinical Relevance: Correlates with age; poor sole predictor of pain

References: Pfirrmann classification spine MRI

RECORD: Centralization vs Peripheralization (McKenzie)

ID: SUP-LUM-004

Topic: Centralization and Peripheralization Phenomenon

Centralization: Distal leg symptoms retreat toward spine with repeated movements — favorable sign

Peripheralization: Symptoms spread distally — stop provocative movement/direction

Application: Guide directional preference therapy; monitor each session

References: McKenzie mechanical diagnosis and therapy

RECORD: Hip-Spine Syndrome

ID: SUP-LUM-005

Topic: Hip-Spine Syndrome — Concurrent Pathology

Concept: Lumbar and hip pathology co-exist; hip ROM loss increases lumbar compensation

Examination: Always screen hip IR/FADIR with LBP especially groin/buttock predominant

Management: Treat both regions; hip arthroplasty may reduce referred LBP

References: Hip-spine syndrome orthopedics literature

24. Evidence and Guidelines

RECORD: NICE NG59 Low Back Pain

ID: EVD-LUM-001

Guideline: NICE NG59 Low Back Pain and Sciatica in Over 16s

Key Recommendations: Reassurance; stay active; avoid routine imaging without red flags; exercise therapy; manual therapy short course; consider CBT for chronic; epidural for sciatica selective

Red Flags: Cauda equina, cancer, infection, fracture, progressive neuro deficit

References: NICE NG59

RECORD: APTA Low Back Pain CPG

ID: EVD-LUM-002

Guideline: APTA Orthopedic Section Low Back Pain Clinical Practice Guideline

Key Recommendations: Classify patients; manual therapy + exercise; trunk stabilization; patient education; psychosocial factors; return to activity emphasis

References: APTA LBP CPG

RECORD: Cochrane Exercise for Chronic LBP

ID: EVD-LUM-003

Review: Cochrane — Exercise therapy for chronic low back pain

Key Findings: Exercise reduces pain and improves function vs no treatment; type less important than doing exercise; moderate certainty

References: Cochrane Library exercise chronic LBP review

RECORD: ACR Appropriateness — Lumbar MRI

ID: EVD-LUM-004

Guideline: ACR Appropriateness Criteria — Low Back Pain

Key Recommendations: MRI appropriate with red flags, progressive neuro deficit, suspected cauda equina, infection, malignancy; usually not first-line uncomplicated acute LBP without red flags; consider after 6 weeks persistent radiculopathy

References: ACR Appropriateness Criteria lumbar spine

RECORD: SPORT Trial — Disc Herniation

ID: EVD-LUM-005

Study: SPORT — Spine Patient Outcomes Research Trial (disc herniation arm)

Key Findings: Surgery (microdiscectomy) provided faster leg pain relief early; differences diminished by 2 years for many patients; shared decision-making emphasized; non-inferiority of conservative care long-term many cases

References: Weinstein et al. SPORT trial publications